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COLLEGE

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Project Report

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Project Report On WATCHERA (E-commerce Website of Watch)

by

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Introduction

4.1 Project Description

WATCHERA is a premium e-commerce platform designed to provide a secure, seamless, and sophisticated online shopping experience for luxury timepieces. By transitioning the elite boutique experience into a responsive digital space, WATCHERA allows customers to confidently browse, evaluate, and purchase high-end watches from anywhere.

Technology Stack

Built on a modern, robust architecture, the platform ensures fast performance and transaction integrity:

- **Frontend:** Angular (delivering a highly responsive, premium user interface).
- **Backend:** Node.js & Express.js (handling secure API routing and CORS configurations).
- **Database:** MongoDB (managing scalable product and user data).

Core Modules

The platform uses a separated module structure to ensure secure and efficient operations across different access levels:

- **Customer Module:** A user-centric interface designed for exploring the timepiece catalog. Features include:
 - Advanced filtering (by brand, style, and price).
 - High-resolution image galleries and detailed technical specifications.
 - User profiles, wishlists, secure carts, and real-time order tracking.
 - Community trust features like user reviews and ratings.
- **Admin Module:** A comprehensive backend dashboard built for platform owners to oversee business operations. Features include:
 - Full database management for products, categories, and users.
 - Sales analytics and financial tracking tools to drive business growth.
 - System configuration capabilities to scale the enterprise.

1.1 Project Profile

Project Title	WATCHERA (E-commerce Website of Watch)
Duration	4 Months
Front-End	Angular , Tailwind CSS
Back-End	Express.js , Node.js , cors
Database	MongoDB
Other Tools	Visual Studio Code, Vite , Microsoft Office, Chrome, Postman , Canva .
Guided By	Prof. Disha Nalwala
Submitted To	SDJ International College, Vesu.

Environment Description

2.1 Hardware and Software Requirements

❖ Hardware Requirement

➤ CLIENT SIDE (Customer's Device):

- Processor: Any device capable of running a modern web browser
- RAM: 4 GB or above
- Storage: 20 GB or above
- Display: Minimum 720p resolution for proper website rendering

➤ DEVELOPMENT SIDE (Developer's Machine):

- Processor: Intel Core i3 / i5 or above
- RAM: 8 GB or above
- Storage: 500 GB or above
- Graphics: Basic graphics card for smooth UI design
- Display: Full HD (1920x1080) preferred

➤ SERVER SIDE (Hosting Environment):

- RAM: 16 GB or above
- Disk Space: 1 TB or above
- Processor: Quad Core or above
- Internet: High-speed connection for handling multiple requests

❖ Software Requirement

➤ CLIENT SIDE (Customer's Browser):

- Operating System: Windows / Linux / macOS / Android / iOS
- Browser: Chrome, Firefox, Safari, Microsoft Edge (latest versions)

➤ DEVELOPMENT SIDE (Developer Tools):

- Operating System: Windows 10 or above
- Front-End: Angular , Tailwind CSS
- Back-End: Node.js ,Express.js , CORS
- IDE/Editor: Visual Studio Code , POSTMAN
- Local Server: MongoDB Atlas ,Local Host
- Documentation Tools: MS Word, MS PowerPoint , Canva

➤ SERVER SIDE (Hosting):

- Web Server: Local Host
- Database: MongoDB
- Browser: Chrome, Internet Explorer, or any modern browser
- Hosting: Shared or Dedicated Hosting with MongoDB Atlas support.

2.2 Technologies Used

❖ Core Technology:

- Front-End : Angular , Tailwind CSS
- Database : MongoDB
- Back-End : Express.js , Node.js , CORS

❖ Extra Technology:

- JavaScript (for interactivity)
- MongoDB Atlas (for database management)
- POSTMAN for API Testing

❖ Documentation Tools:

- Microsoft Word (for report writing)
- Microsoft PowerPoint (for presentations)
- Canva for Diagram , Graphs , charts

❖ Development Tools:

- Visual Studio Code
- Postman
- Chrome Browser (testing & debugging)

Technologies in Detail

1) Visual Studio Code (VS Code):

VS Code is a lightweight, open-source code editor by Microsoft used to build the WATCHERA platform. It accelerates development with smart code suggestions (IntelliSense), an integrated terminal for executing Angular CLI and Node.js commands, and built-in Git for version control. Its vast library of extensions and Live Share features make writing, debugging, and collaborating on code highly efficient.

2) Angular :

Angular is a robust, TypeScript-based framework by Google used to build WATCHERA's scalable Single-Page Application (SPA). Its modular, component-based architecture and two-way data binding make maintaining complex user interfaces highly efficient. With built-in routing and secure HTTP tools, it flawlessly handles dynamic product catalogs, secure shopping carts, and multi-role dashboards (Customer, Staff, Admin).

3) Tailwind CSS:

Tailwind CSS is a utility-first framework used to rapidly design WATCHERA's visually sophisticated interface. By applying pre-defined utility classes directly in the HTML, developers can ensure a consistent, premium aesthetic that perfectly matches the brand's luxury guidelines. Its built-in responsive design guarantees a flawless, mobile-first shopping experience across smartphones, tablets, and desktops..

4) Node.js & Express.js:

Node.js is an asynchronous, event-driven JavaScript runtime that enables server-side execution. Paired with Express.js—a fast, minimalist web framework—they form the high-performance backend engine for WATCHERA. Together, they handle secure RESTful API routing, seamlessly connecting the Angular frontend to the database while ensuring rapid response times.

5) CORS :

CORS is a crucial HTTP-header-based security mechanism implemented by web browsers. It allows the Node.js backend to explicitly and securely accept data requests from the Angular frontend running on a different domain. This ensures safe, reliable communication while protecting user data and transaction integrity.

6) MongoDB:

MongoDB is a scalable NoSQL database that stores data in flexible, JSON-like documents. This dynamic, schema-less structure is perfect for WATCHERA's diverse catalog, easily accommodating the varying technical specifications of different luxury timepieces. It ensures high-speed read/write operations for complex queries, securely managing user profiles, shopping carts, and order histories.

Platform Advantages Summary

- **Scalability & Speed:** The combination of MongoDB's distributed database and Angular's SPA rendering easily handles high traffic and delivers a fluid, app-like experience.
- **Unified Workflow:** Utilizing a full JavaScript/TypeScript (MEAN) stack creates a highly efficient, seamless pipeline between the backend APIs and frontend components.
- **Cost-Effective Security:** Leveraging a community-backed open-source ecosystem provides enterprise-level reliability and security without prohibitive licensing costs.

System Analysis and Planning

3.1 Existing System and its Drawbacks

Currently, the luxury watch industry relies on restrictive brick-and-mortar boutiques and outdated, generic e-commerce platforms. Both models present significant limitations that hinder business growth and customer satisfaction:

- **Geographical & Time Limits:** Physical stores restrict sales to local areas and specific operating hours, severely limiting market reach.
- **Inadequate Role Segregation:** Legacy software often lacks a dedicated "Staff" module. Store owners are forced to either handle all operations themselves or risk security by giving regular employees full Admin access.
- **Clunky UX & Performance:** Traditional multi-page websites suffer from slow load times and poor mobile optimization, breaking the immersive feel expected by high-end buyers.
- **Rigid Inventory Management:** Outdated relational databases struggle to adapt. Adding new watch types with entirely different technical specs (e.g., smartwatches vs. mechanical) requires time-consuming database restructuring.
- **Generic Aesthetics:** Standard e-commerce templates fail to capture the exclusivity and sophisticated branding required to sell luxury investments.

Conclusion: There is a clear need for a scalable, high-performance platform like **WATCHERA**. By utilizing a modern Single Page Application (SPA) architecture and flexible NoSQL database, WATCHERA bridges the gap between premium luxury retail and modern web convenience.

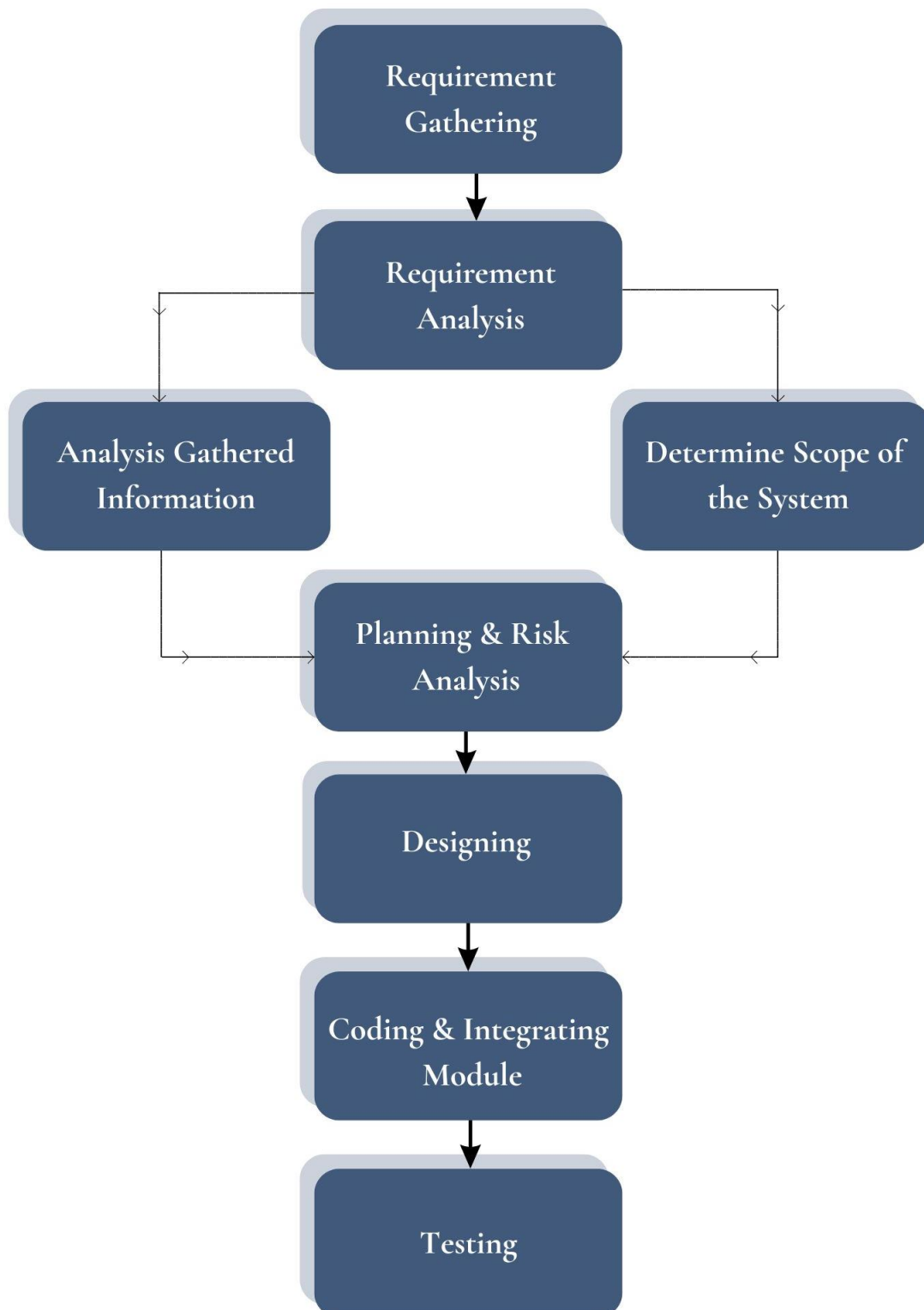
3.2 Feasibility Study

This study confirms that the WATCHERA platform is technically possible, economically justifiable, and operationally sound before committing extensive development resources.

- **1. Technical Feasibility:** * Utilizes the proven MEAN stack (MongoDB, Express.js, Angular, Node.js) and Tailwind CSS.
 - Angular's Single Page Application (SPA) architecture ensures the fast, interactive experience required for a premium store.
 - Standardized REST API integrations guarantee stable connections to databases and secure payment gateways.
- **2. Economic Feasibility:** * Leveraging open-source technologies completely eliminates software licensing fees.
 - Cloud-based hosting (e.g., MongoDB Atlas) provides a highly cost-effective, pay-as-you-go scaling model.
 - Operating a 24/7 digital storefront eliminates physical overhead, resulting in a highly favorable ROI for high-margin luxury timepieces.
- **3. Operational Feasibility:** * Distinct Customer, Staff, and Admin modules securely streamline daily workflows and resolve legacy bottlenecks.
 - An intuitive, Tailwind-designed UI ensures a minimal learning curve for both shoppers and backend employees.
 - Automated order tracking, inventory updates, and secure data handling significantly reduce manual administrative work.

Conclusion: The WATCHERA project is highly feasible across all parameters. The open-source stack mitigates financial risk, the architectural design guarantees technical stability, and the role-based system ensures smooth daily operations.

❖ Task Dependency



3.3 Requirement Gathering and Analysis

This phase establishes the blueprint for the WATCHERA platform, ensuring the technology stack accurately addresses both luxury market expectations and operational business goals.

1. Functional Requirements

These define the specific features the system must perform, categorized by user access levels:

Customer Module:

- **Discovery & Purchase:** Advanced search filters, high-resolution imagery, and secure cart/checkout functionality.
- **Account Management:** Secure registration, personal profiles, wishlists, order history tracking, and product reviews.
- **Staff Module: Daily Operations:** Processing incoming orders, updating shipping statuses, and monitoring inventory to prevent overselling.
- **Customer Service:** Managing basic customer support queries and return requests.

Admin Module:

- **Platform Control:** Full CRUD (Create, Read, Update, Delete) capabilities for the master product catalog and categories.
- **Business Oversight:** Complete user management (Customers and Staff) and access to comprehensive sales analytics and revenue reports.

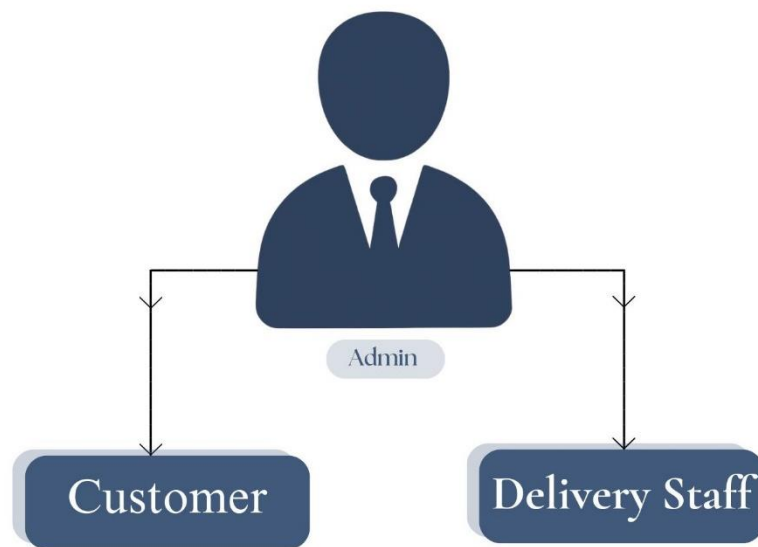
2. Non-Functional Requirements

These define the quality, performance, and security standards of the platform:

- **Security:** Robust data protection utilizing JWT (JSON Web Tokens) for secure sessions, strict CORS configurations, and bcrypt password encryption within the MongoDB database.
- **Performance:** Guaranteed fast load times (under 3 seconds)—even during high-traffic periods with high-resolution images—powered by Angular's efficient rendering and Node.js's asynchronous processing.

Proposed System

4.1 Scope



❖ Admin / Owner:

Serving as the platform's central command center, the Admin holds the highest level of authorization with absolute control over backend operations, inventory, and users:

1. **Dashboard & Analytics:** Real-time visibility into vital business metrics, including total revenue, sales volume, and customer acquisition.
2. **Catalog Management (CRUD):** Full authority to add, edit, or delete product listings, upload high-resolution images, and define technical specifications.
3. **Order & Revenue Management:** Complete oversight of the order lifecycle, including payment verification, refunds, shipping tracking, and invoicing.
4. **User & Access Control:** Manage all registered accounts and securely create restricted "Staff" roles to delegate daily tasks without exposing sensitive financial data.
5. **Marketing & Promotions:** Configure platform-wide banners, generate discount codes, and curate featured or "limited edition" timepiece collections.

❖ Customer:

Focused on delivering a frictionless, premium shopping journey, this module restricts users to frontend interactions and personal data management:

1. **Discovery & Filtering:** Seamlessly browse the luxury catalog using advanced, dynamic filters (brand, price, movement) for instant results without page reloads.
2. **Profile Management:** Securely register accounts, manage encrypted credentials, and store multiple shipping or billing addresses.
3. **Cart & Checkout:** Manage a persistent shopping cart, apply promo codes, and complete purchases via a secure, multi-step payment gateway.
4. **Wishlists & Reviews:** Curate personal wishlists for future consideration and submit post-purchase ratings to build community trust.
5. **Order Tracking:** Access a dedicated portal to view complete purchase histories, download invoices, and track active orders in real-time.

4.2 Project Modules

- 1. User Registration & Authentication** Secure sign-up and login via email, phone, or social accounts. Features bcrypt-encrypted passwords, automated OTP recovery, and comprehensive profile management for storing multiple secure billing and shipping addresses.
- 2. Product Search & Catalog** Dynamic browsing of curated luxury collections with advanced filtering (by brand, price, movement type, and material). Premium product pages display high-resolution galleries, detailed technical specs, and warranty information.
- 3. Shopping Cart & Wishlist** Customers can curate a persistent wishlist and manage a dynamic, reload-free shopping cart. The checkout process provides transparent, real-time summaries detailing base prices, luxury taxes, and insured shipping costs.
- 4. Payment & Billing** Integrates secure payment gateways (cards, wallets, UPI) for high-value transactions. Supports promotional discount codes, generates automated digital invoices (including serial numbers), and maintains a complete transaction history.
- 5. Order & Fulfillment Management** Customers can track their orders in real-time. Staff members can manage order queues, update fulfillment statuses (e.g., "Quality Check," "Shipped"), generate digital packing slips, and input secure courier tracking numbers.
- 6. Product Management (Admin)** Grants Admins full CRUD authority over the catalog and media uploads. Includes real-time inventory synchronization to prevent overselling, dynamic pricing controls, and overarching financial oversight to manage revenue and refunds.
- 7. Dashboard & Analytics (Admin & Staff)** Admins can manage user access roles and view comprehensive business intelligence (sales, traffic, and revenue growth). Staff utilize their dashboard to monitor daily active orders and resolve low-stock warnings.
- 8. Review & Rating** Authenticated customers can submit verified star ratings and written reviews, which are dynamically displayed on product pages to build community trust. Staff and Admins can actively monitor and moderate this feedback.
- 9. Customer Support** Features a detailed self-service FAQ (warranties, watch care) and direct assistance via live chat or inquiry forms. A dedicated ticketing system empowers Staff to efficiently track and resolve customer complaints or return requests.

4.3 Module-Wise Objectives, Functionalities, and Constraints

This section outlines the core modules of the WATCHERA platform, detailing their primary goals, key features, and system limitations to ensure smooth operation and maintain a premium user experience.

1. User Authentication & Profile Management

- **Objective:** To provide secure, seamless access and a personalized experience for customers.
- **Functionalities:** User registration, secure login/logout, profile detail updates, order history tracking, and wishlist management.
- **Constraints:** Passwords must be strictly encrypted before database storage. User sessions automatically expire after 24 hours of inactivity for security. Password reset requests require email verification.

2. Product Catalog & Search

- **Objective:** To enable intuitive and efficient discovery of luxury timepieces.
- **Functionalities:** Dynamic product grid rendering, advanced filtering (by brand, price, movement, gender), keyword search, and detailed product view pages.
- **Constraints:** High-resolution image uploads are restricted to a maximum of 5MB per file to maintain page load speeds. Search queries are capped at 50 characters to optimize MongoDB performance. Pagination is strictly enforced at 20 items per page.

3. Cart & Order Processing

- **Objective:** To facilitate a frictionless, error-free checkout and transaction process.
- **Functionalities:** Real-time cart updates, coupon code validation, secure payment gateway integration, order confirmation, and automated invoice generation.
- **Constraints:** Only one discount coupon can be applied per order (no stacking). Shopping cart sessions expire after 7 days of inactivity. Payment processing is dependent on the uptime and latency of the integrated third-party payment gateway.

4. Admin Dashboard & Inventory Management

- **Objective:** To provide a centralized, powerful interface for store administrators to manage the platform.
- **Functionalities:** Complete CRUD (Create, Read, Update, Delete) operations for products and users, stock level monitoring, order status management, and sales reporting.
- **Constraints:** Access is strictly limited to accounts with verified "Admin" role privileges. To prevent database overload, real-time sales and traffic reports are cached and updated in 5-minute intervals rather than continuously.

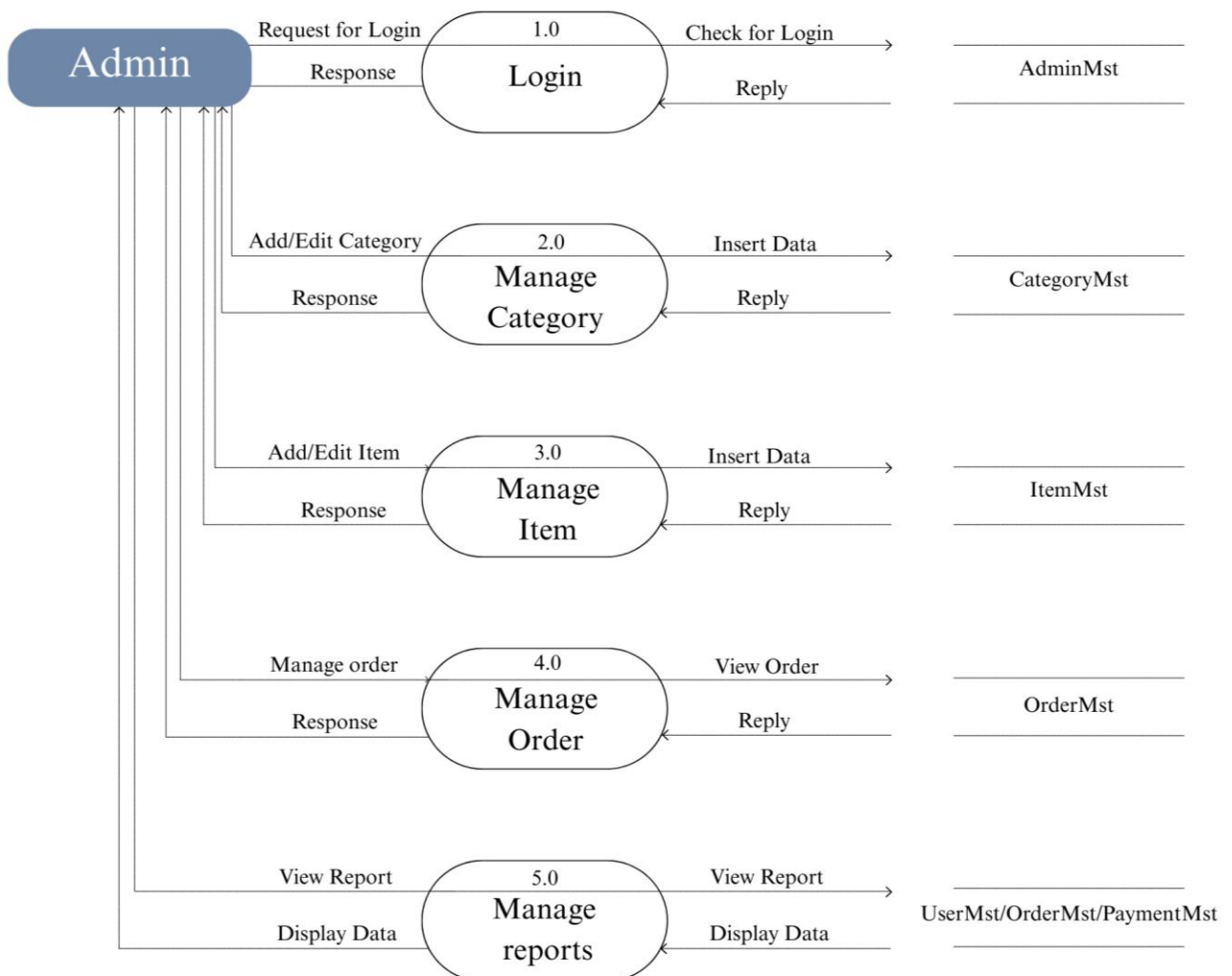
Detail Planning

5.1 Data Flow Diagram

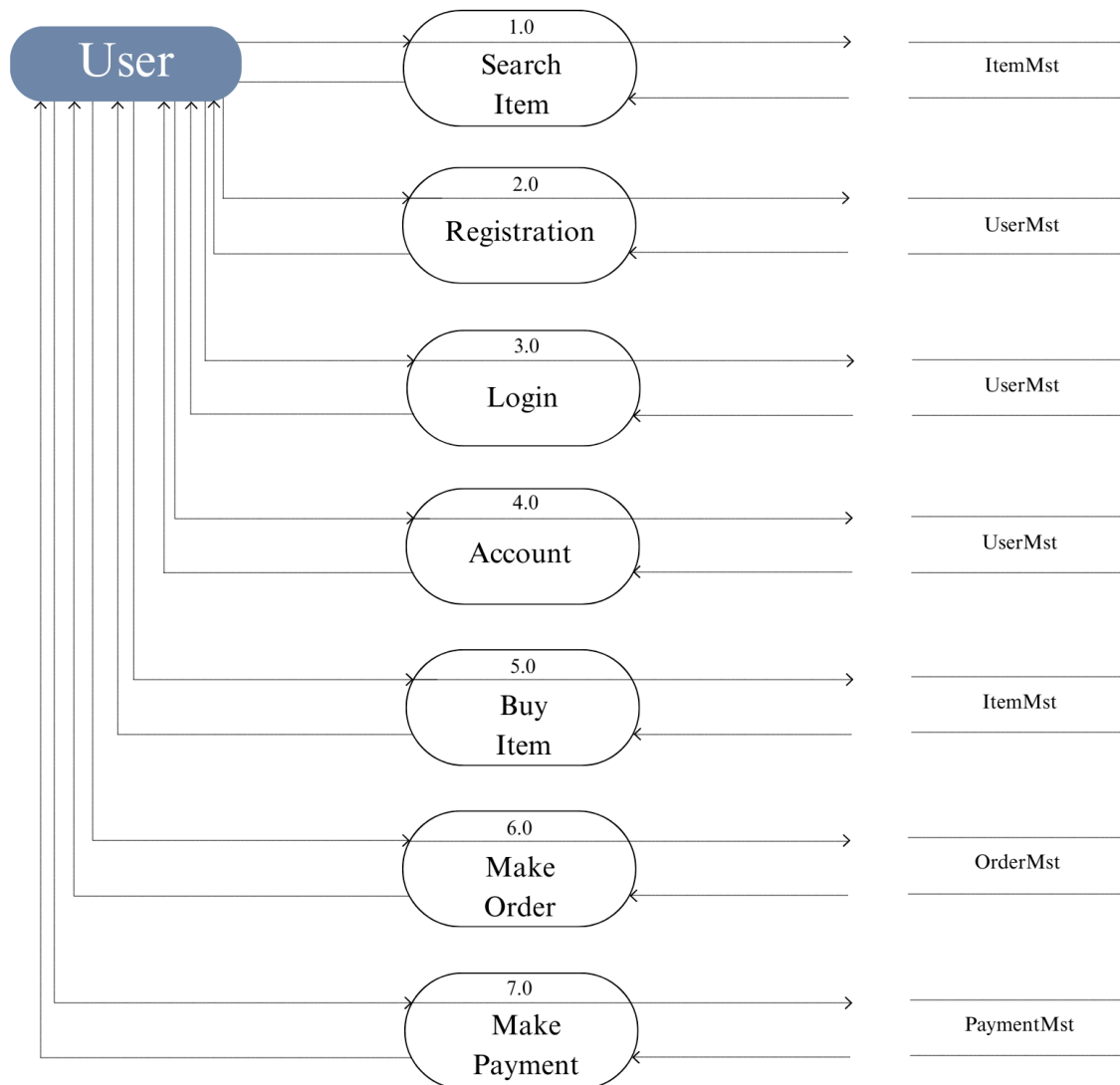
Context-level



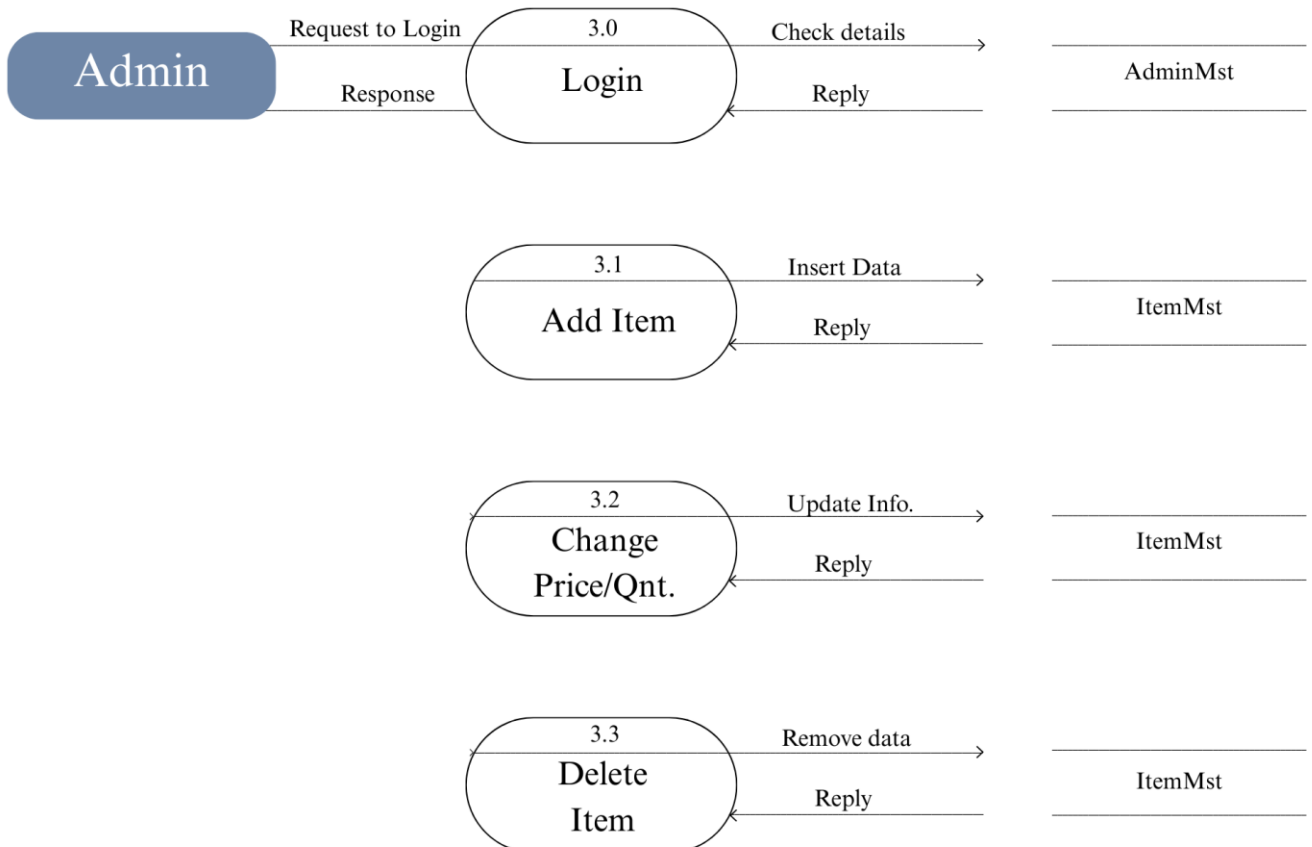
Admin side DFD - 1st level



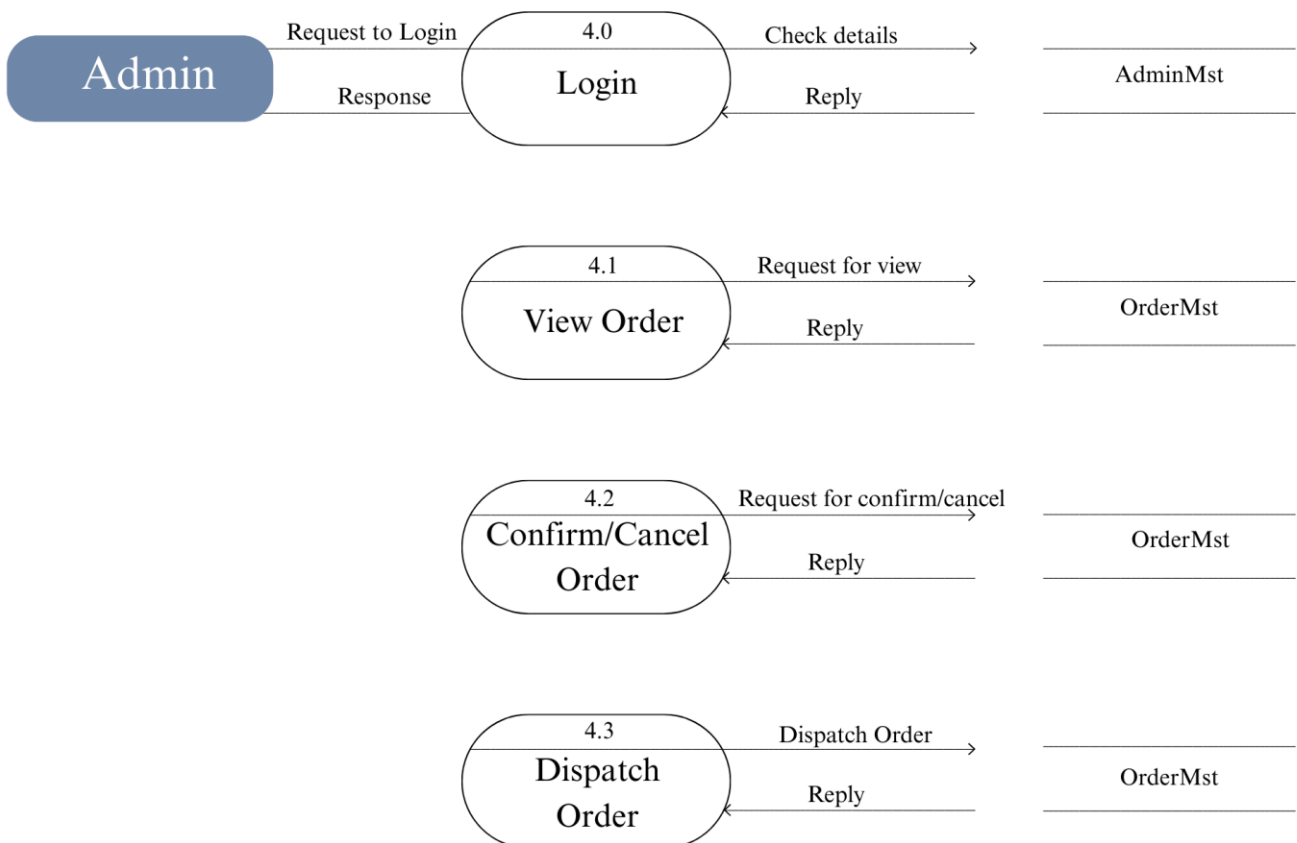
User side DFD - 1st level



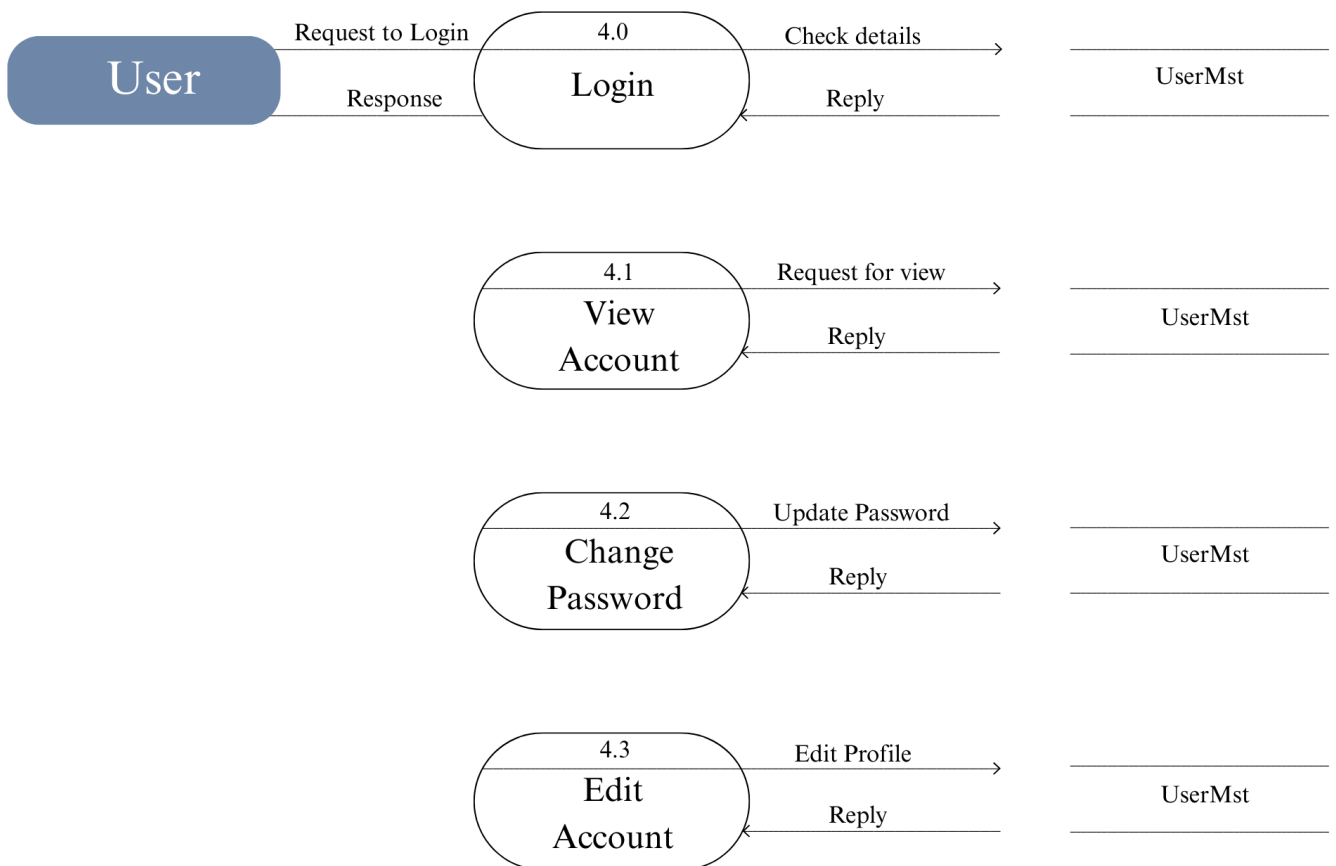
Admin side DFD - 2nd level [3.0]



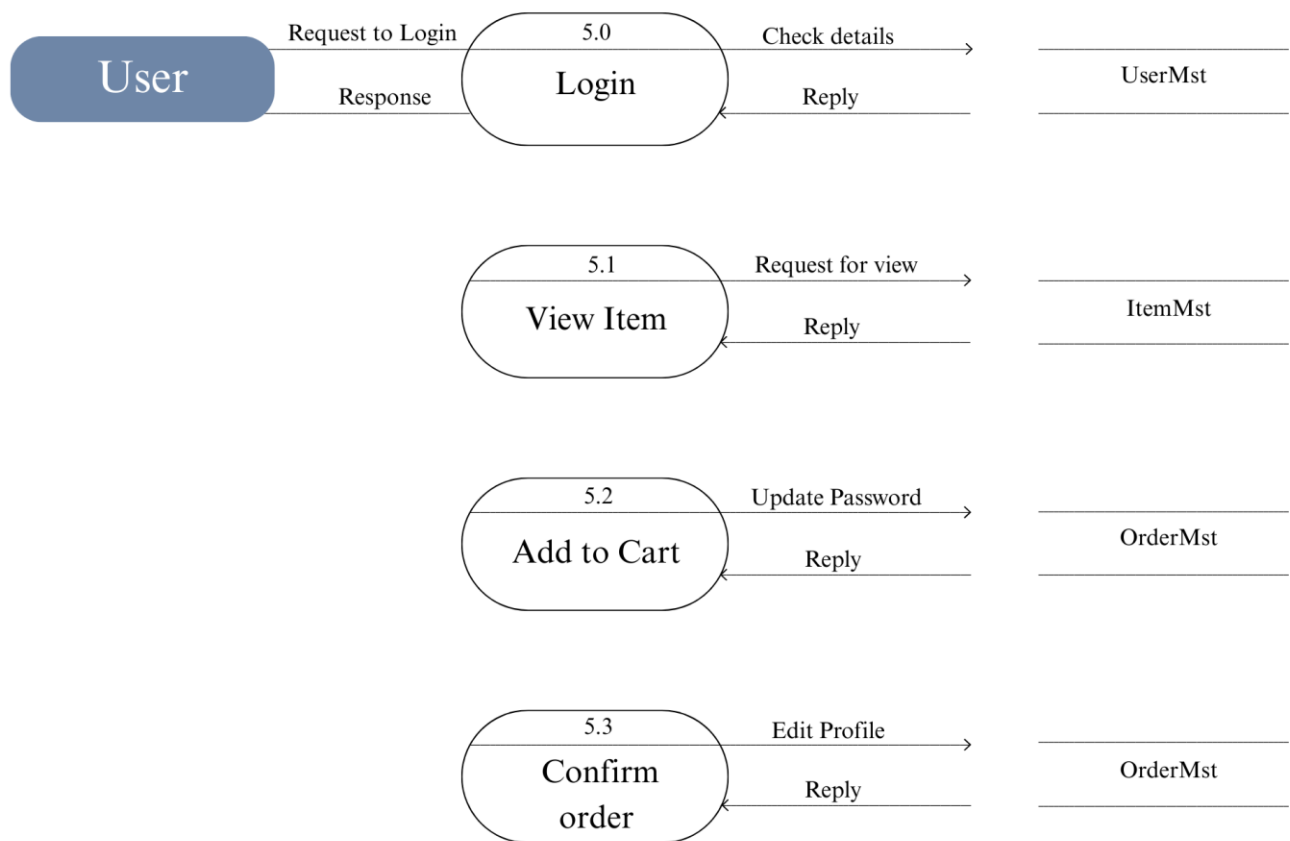
Admin side DFD - 2nd level [4.0]



User side DFD - 2nd level [4.0]

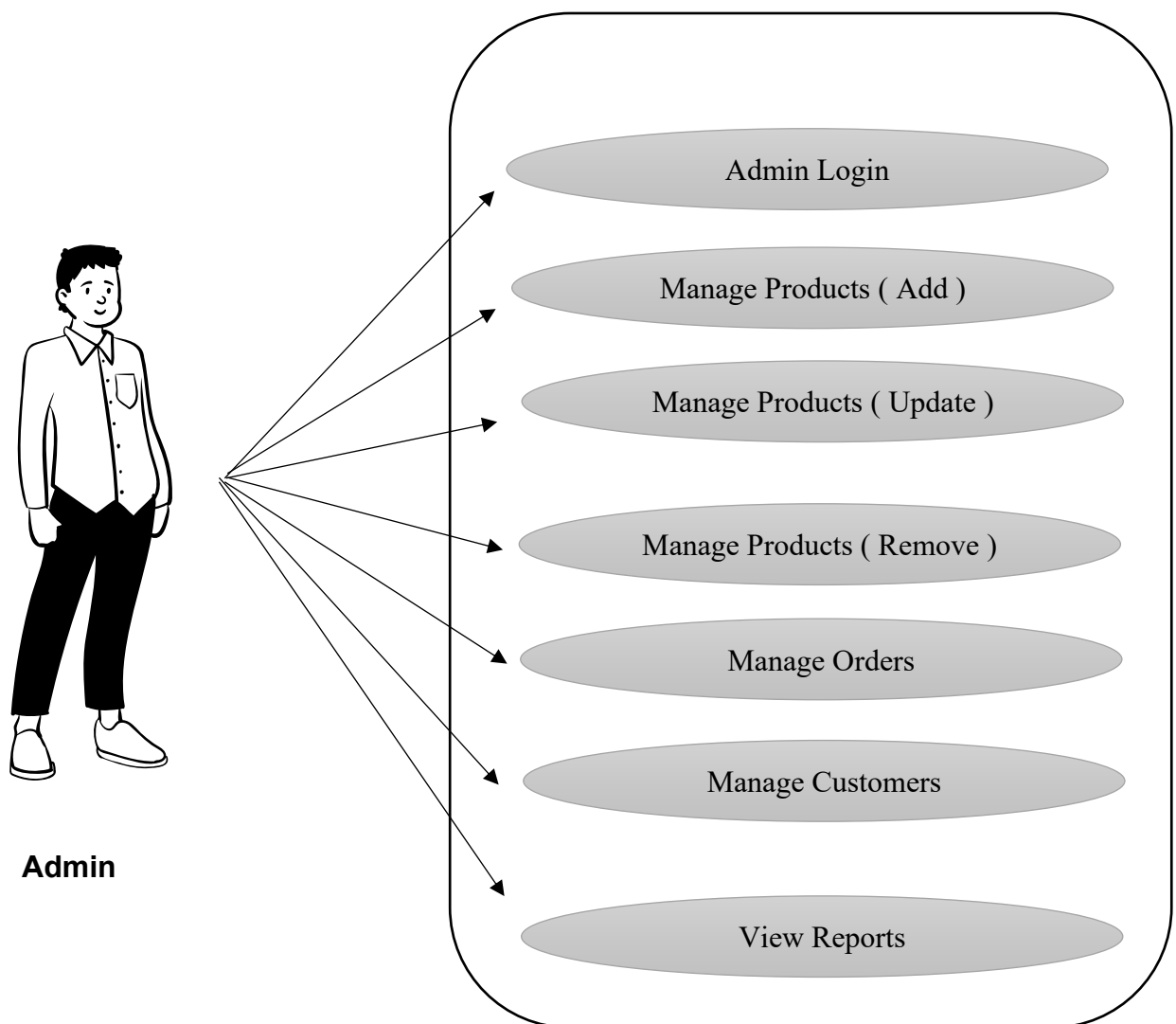


User side DFD - 2nd level [5.0]

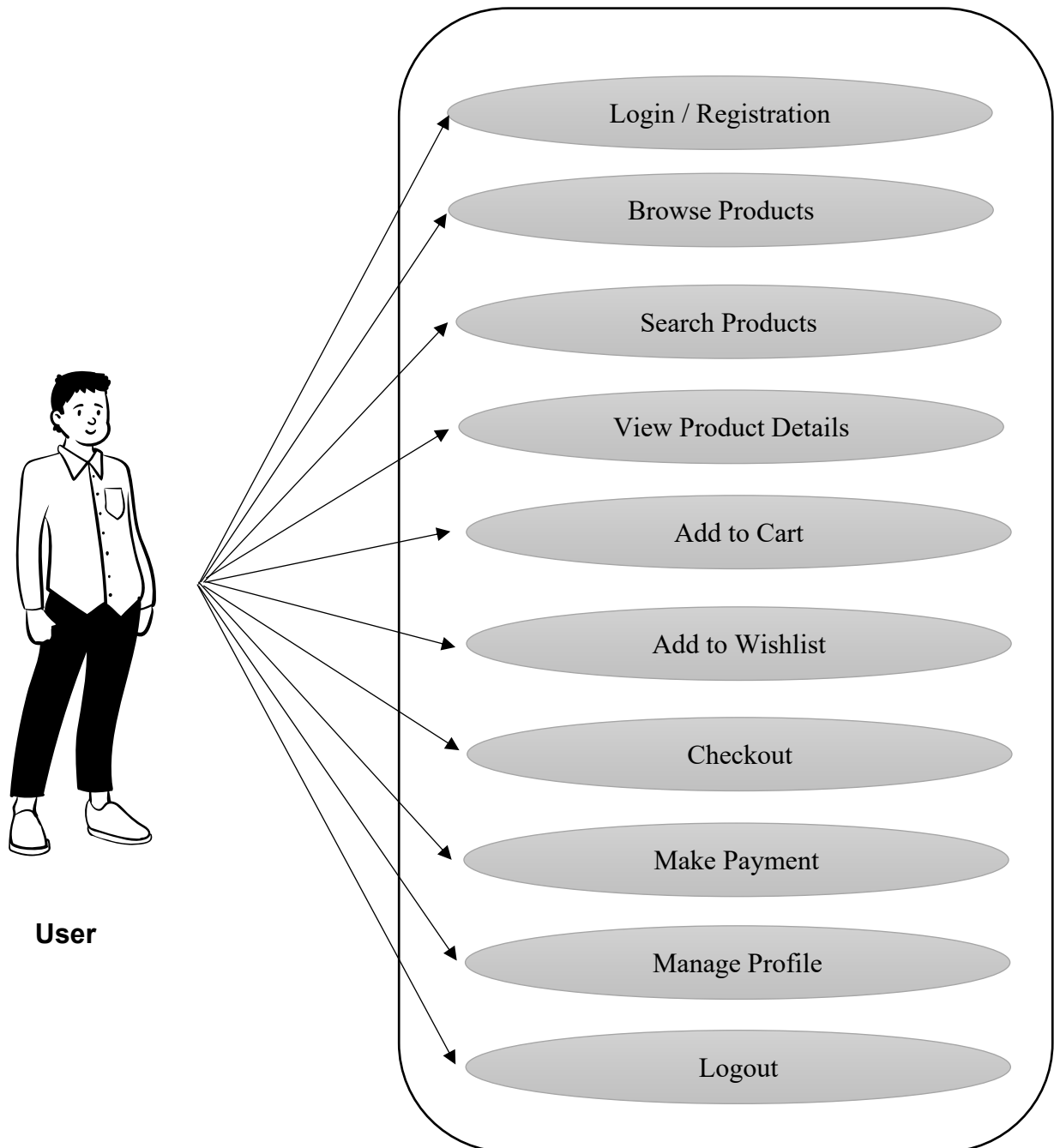


5.2 UseCase Diagram

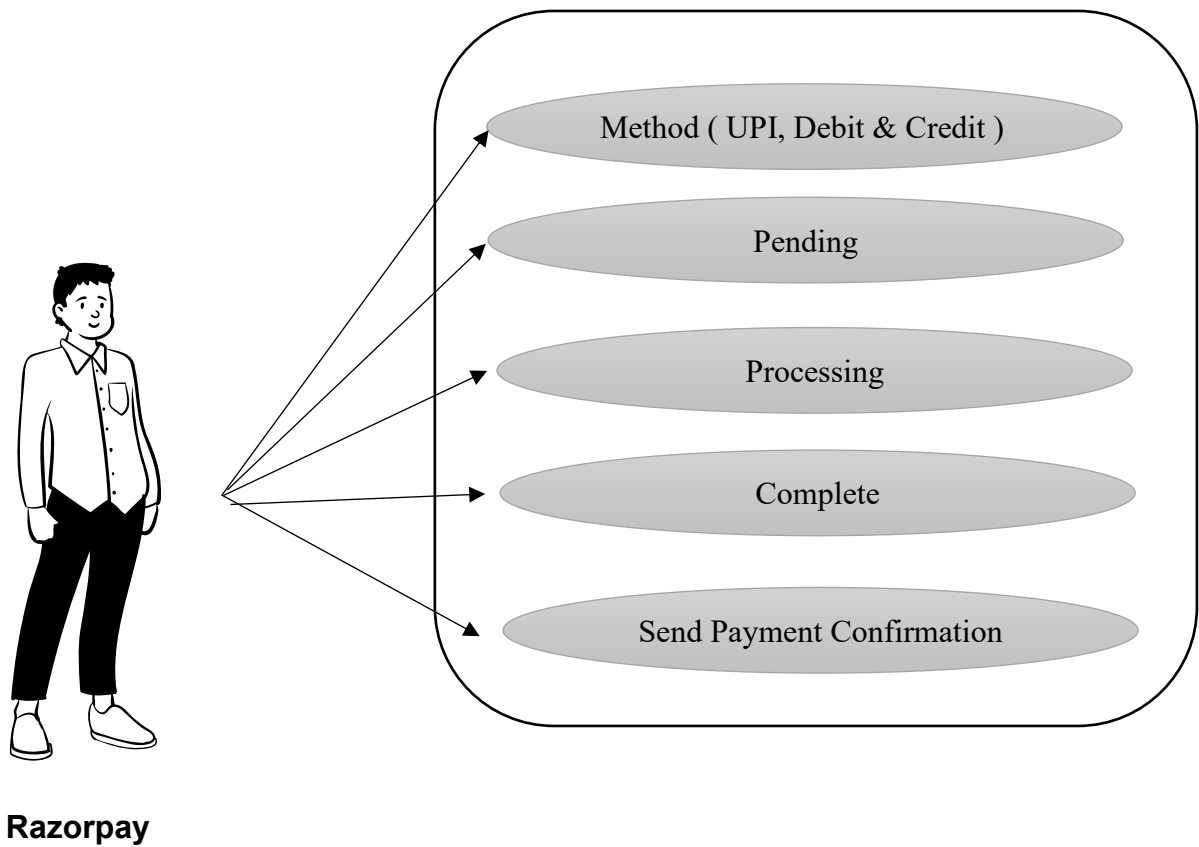
❖ Admin Use Case Diagram



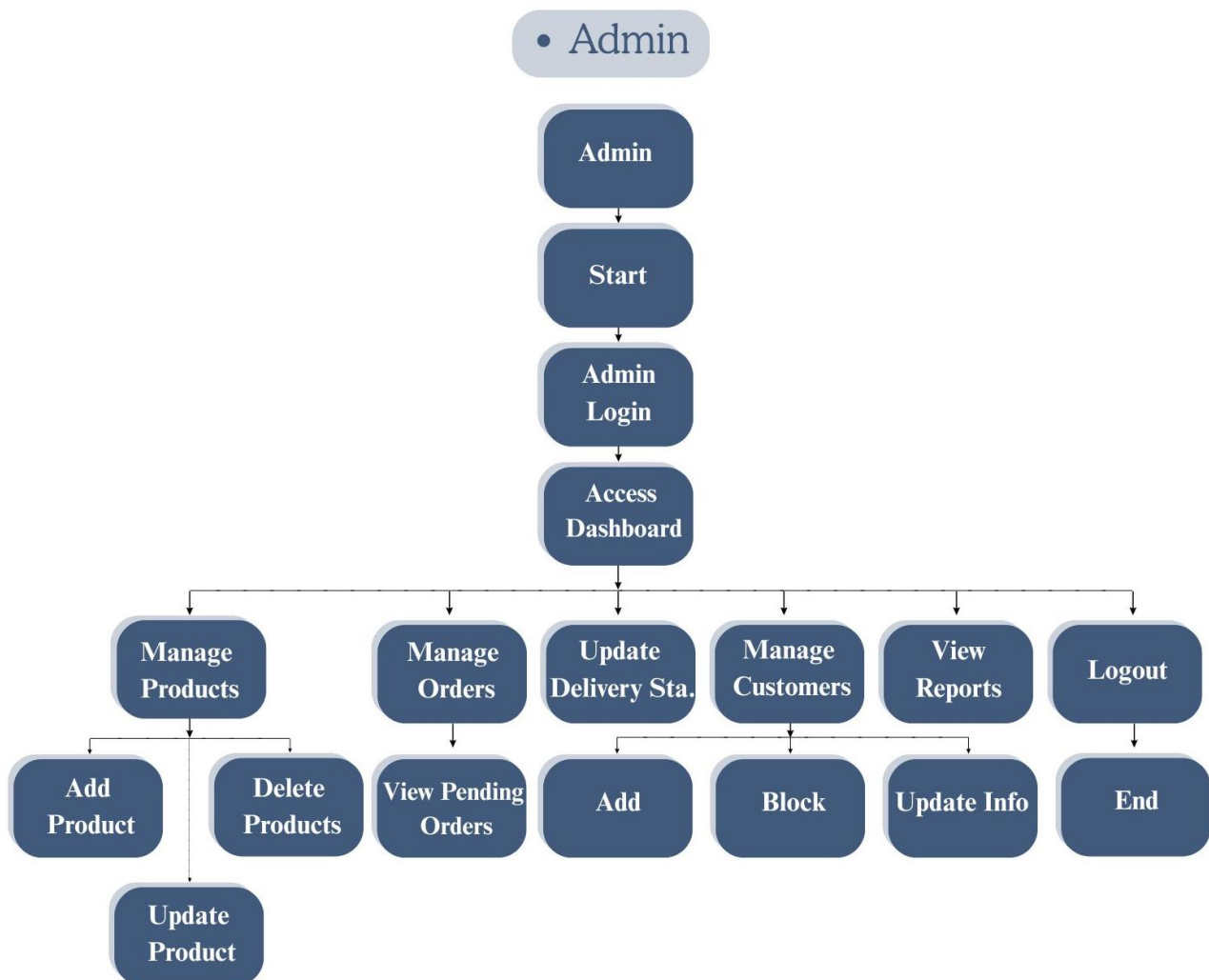
❖ Customer Use Case Diagram

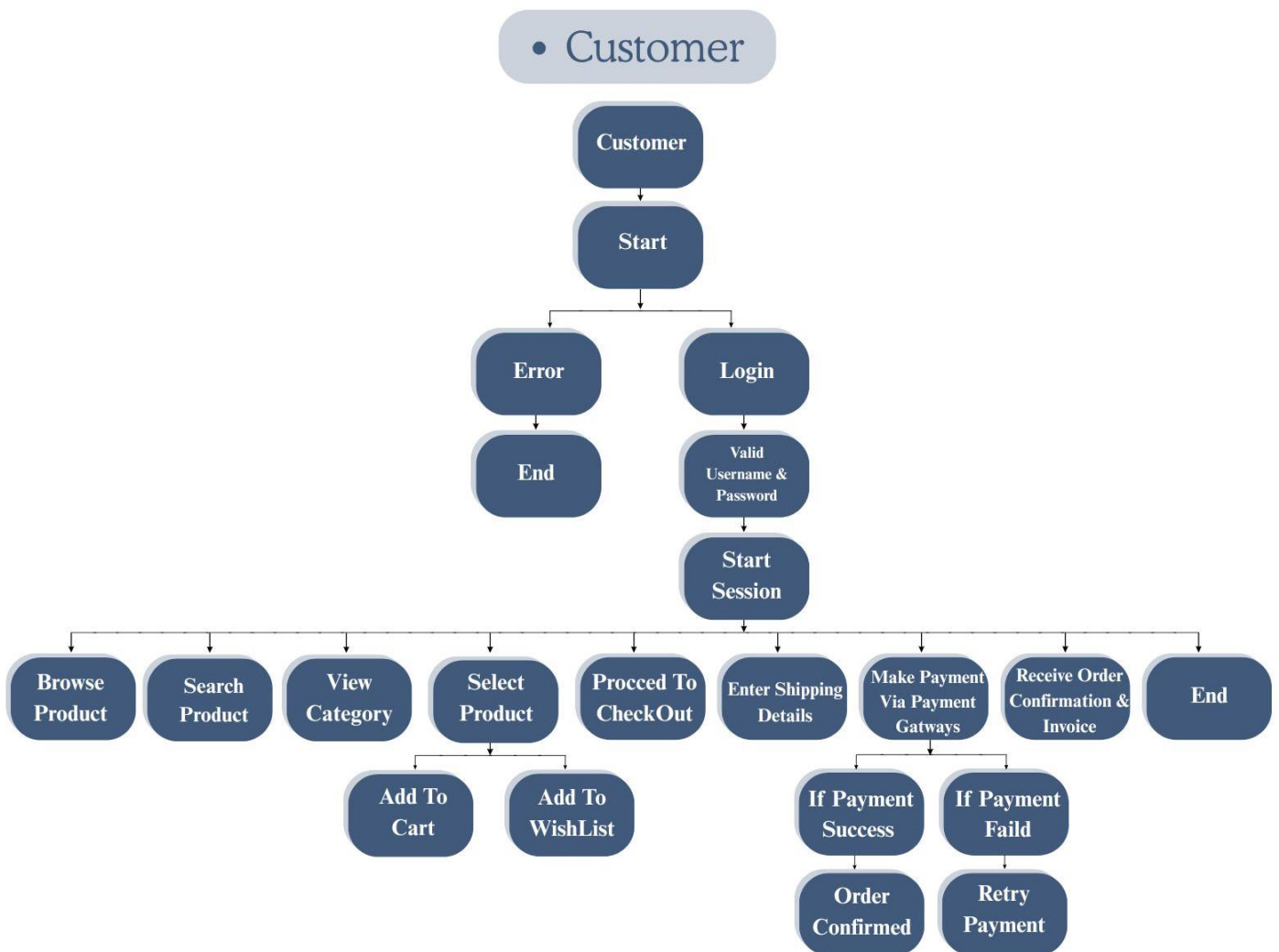


❖ Razorpay Payment Diagram

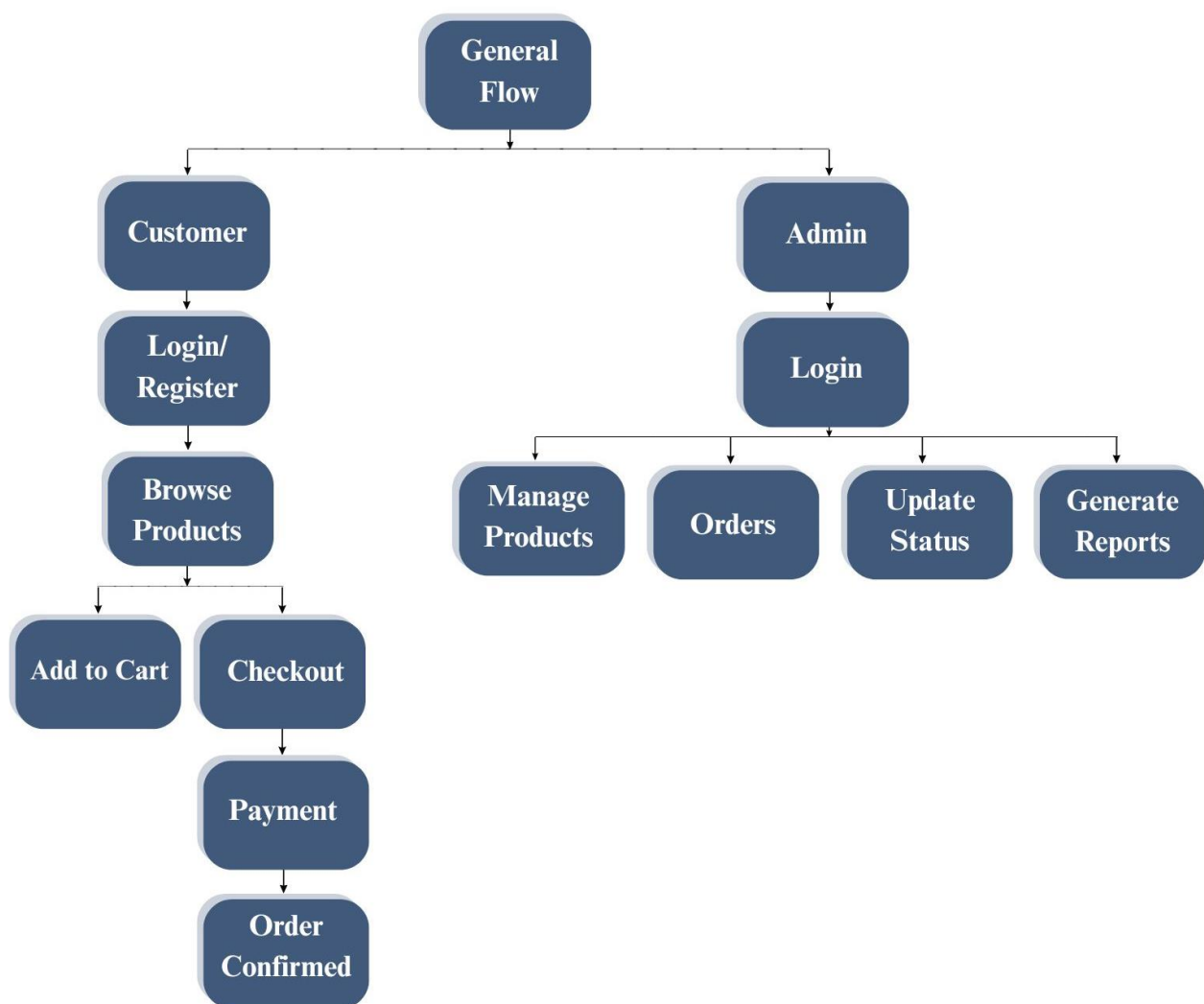


5.3 Activity Flow Diagram





- General Flow Representation



System Design

6.1 Database Design

❖ User Table

Field Name	Description	Data Type
_id	Unique identifier for each user	ObjectId
name	Full name of the user	String
email	Email address of the user	String
password	Encrypted password for authentication	String
role	Defines user role (Admin / User)	String
isActive	Indicates whether the account is active	Boolean

❖ Cart Table

Field Name	Description	Data Type
_id	Unique cart identifier	ObjectId
user	Reference to the user	ObjectId
items	List of products added to cart	Array
product	Product reference	ObjectId
quantity	Quantity of product in cart	Number
createdAt	Cart creation date	Date
updatedAt	Cart last update date	Date

❖ **Wishlist Table**

Field Name	Description	Data Type
_id	Unique wishlist identifier	ObjectId
user	Reference to user	ObjectId
products	List of saved products	Array
createdAt	Wishlist creation date	Date
updatedAt	Wishlist last update date	Date

❖ **Coupons Table**

Field Name	Description	Data Type
_id	Unique coupon identifier	ObjectId
code	Coupon code	String
discountType	Type of discount (percentage/fixed)	String
discountValue	Discount value	Number
minPurchase	Minimum purchase required	Number
expiryDate	Coupon expiration date	Date
usageLimit	Maximum usage limit	Number
usedCount	Number of times coupon used	Number
isActive	Indicates whether coupon is active	Boolean
createdAt	Coupon creation date	Date
updatedAt	Coupon last update	Date

❖ **Shipping Zones Table**

Field Name	Description	Data Type
_id	Unique identifier for shipping zone	ObjectId
zoneName	Name of delivery zone	String
shippingCost	Delivery cost for the zone	Number
estimatedDelivery	Estimated delivery time	String
isActive	Indicates zone availability	Boolean

❖ **Product Table**

Field Name	Description	Data Type
_id	Unique identifier of the product	ObjectId
name	Name of the watch product	String
slug	URL friendly product name	String
description	Detailed description of the product	String
shortDescription	Short summary of the product	String
brand	Brand name of the watch	String
modelNumber	Unique model number	String
sku	Stock keeping unit	String
category	Product category	String
subcategory	Product subcategory	String
price	Original price of the product	Number
discountPrice	Discounted selling price	Number
stock	Available product quantity	Number
images	Product images	Array
video	Product video link	String
movement	Watch movement type	String
caseMaterial	Material of watch case	String
strapMaterial	Material of strap	String
dialColor	Color of watch dial	String
caseSize	Size of the watch case	String
waterResistance	Water resistance capacity	String
warranty	Warranty period	String
isFeatured	Indicates featured product	Boolean
isNewArrival	Indicates new arrival product	Boolean
isBestSeller	Indicates bestseller product	Boolean
isActive	Indicates whether product is available	Boolean
createdAt	Product creation date	Date
updatedAt	Last update date	Date

❖ **Order Table**

Field Name	Description	Data Type
_id	Unique identifier of the order	ObjectId
user	Reference to the user placing the order	ObjectId
orderItems	List of products in the order	Array
shippingAddress	Address for product delivery	Object
paymentMethod	Payment mode used by customer	String
itemsPrice	Total price of items	Number
taxPrice	Tax amount applied	Number
shippingPrice	Shipping charges	Number
totalPrice	Final payable amount	Number
isPaid	Indicates if payment is completed	Boolean
isDelivered	Indicates if order is delivered	Boolean
orderStatus	Current order status	String
couponDiscount	Discount applied from coupon	Number
createdAt	Order creation date	Date
updatedAt	Last order update	Date
deliveredAt	Delivery completion date	Date

6.2 User Interface

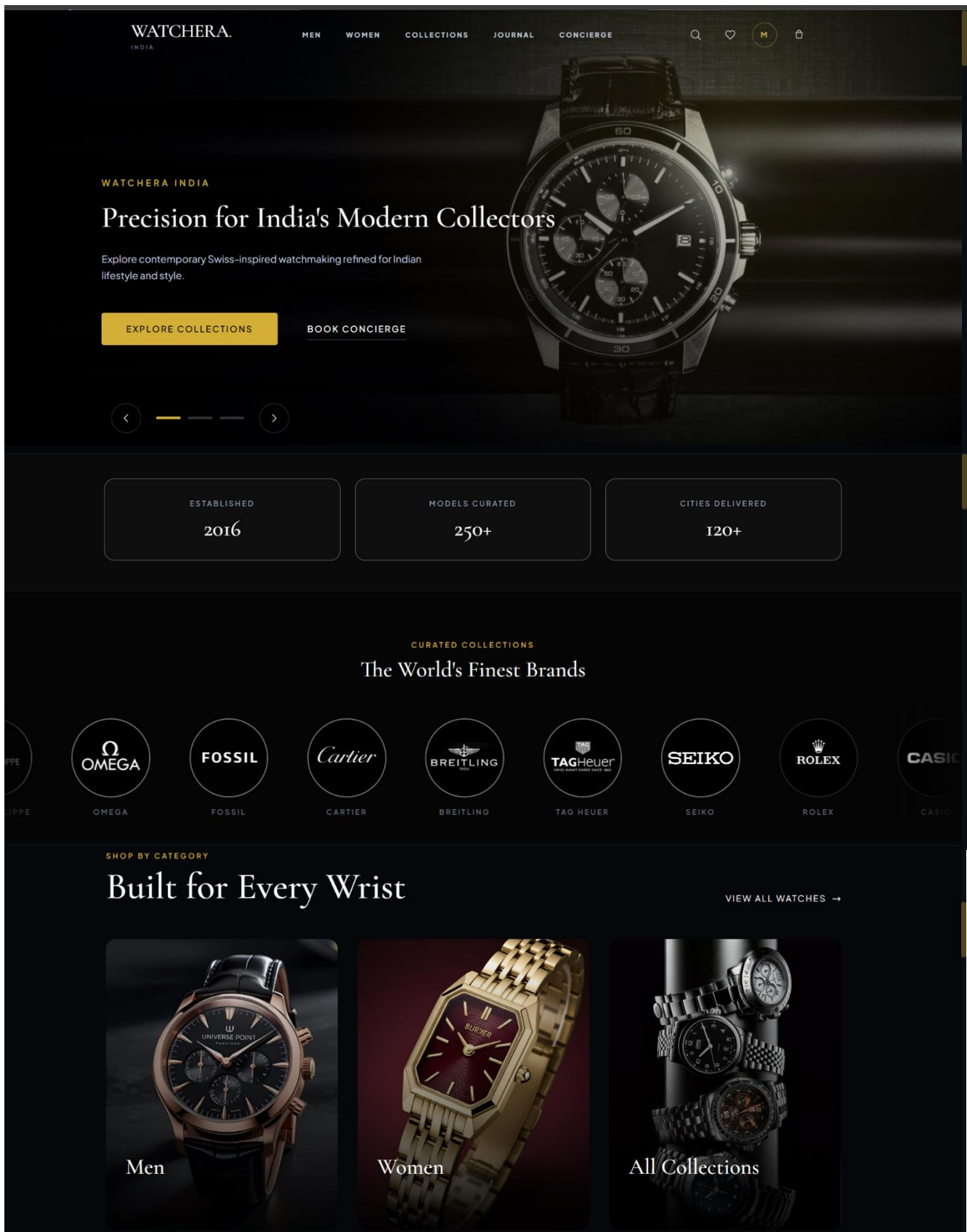
The User Interface (UI) of WATCHERA is designed to immediately convey a sense of luxury, exclusivity, and technological sophistication. Built utilizing Angular components and styled entirely with Tailwind CSS, the platform features a premium "dark mode" aesthetic accented with elegant gold and crisp white typography. The layout is fully responsive, ensuring a seamless experience across all devices.

Home Page Layout & Components

The digital storefront is strategically laid out to guide customers from visual impact to purchase across nine key sections:

- **Global Navigation Bar:** Clean branding, primary category links, and quick-action icons (search, cart, profile).
- **Dynamic Hero Section:** Striking, full-width watch imagery featuring rotating campaigns and distinct "Shop Now" / "Learn More" calls-to-action (CTAs).
- **Platform Statistics Bar:** A data ribbon displaying metrics like brand count and customer base to build immediate credibility.
- **Category Discovery:** Visual cards directing users to "Men," "Women," and "Collections."
- **Core Value Propositions:** Trust-building highlights focusing on verified authenticity, secure nationwide delivery, and concierge support.
- **Dynamic Product Carousels:** MongoDB-powered sliders displaying "New Arrivals" and "Best Sellers" with high-quality images and clear pricing.
- **Educational Content:** Buyer's guides and articles aimed at assisting first-time luxury watch purchasers.
- **Support Call-to-Action:** A dedicated section offering premium, one-on-one chat assistance for high-value purchases.
- **Comprehensive Footer:** Essential quick links, brand summaries, and a newsletter subscription input.

Home Page



The mockup displays the WATCHERA India website home page. At the top, a navigation bar includes the brand name, category links (MEN, WOMEN, COLLECTIONS, JOURNAL, CONCIERGE), and utility icons (search, heart, menu, cart). The main hero section features a large image of a chronograph watch and the headline "Precision for India's Modern Collectors", followed by a sub-headline and two call-to-action buttons: "EXPLORE COLLECTIONS" and "BOOK CONCIERGE". Below this is a statistics row showing "ESTABLISHED 2016", "MODELS CURATED 250+", and "CITIES DELIVERED 120+". The "CURATED COLLECTIONS" section lists eight brands: Patek Philippe, Omega, Fossil, Cartier, Breitling, Tag Heuer, Seiko, and Rolex, each with a circular logo and brand name. The "SHOP BY CATEGORY" section is titled "Built for Every Wrist" and includes a "VIEW ALL WATCHES" link. It features three category tiles: "Men" with a chronograph watch, "Women" with a rectangular watch, and "All Collections" with a stack of various watches.

WATCHERA.
INDIA

MEN WOMEN COLLECTIONS JOURNAL CONCIERGE

WATCHERA INDIA

Precision for India's Modern Collectors

Explore contemporary Swiss-inspired watchmaking refined for Indian lifestyle and style.

EXPLORE COLLECTIONS BOOK CONCIERGE

ESTABLISHED 2016

MODELS CURATED 250+

CITIES DELIVERED 120+

CURATED COLLECTIONS

The World's Finest Brands

Patek Philippe OMEGA FOSSIL Cartier BREITLING TAG Heuer SEIKO ROLEX CASIO

SHOP BY CATEGORY

Built for Every Wrist

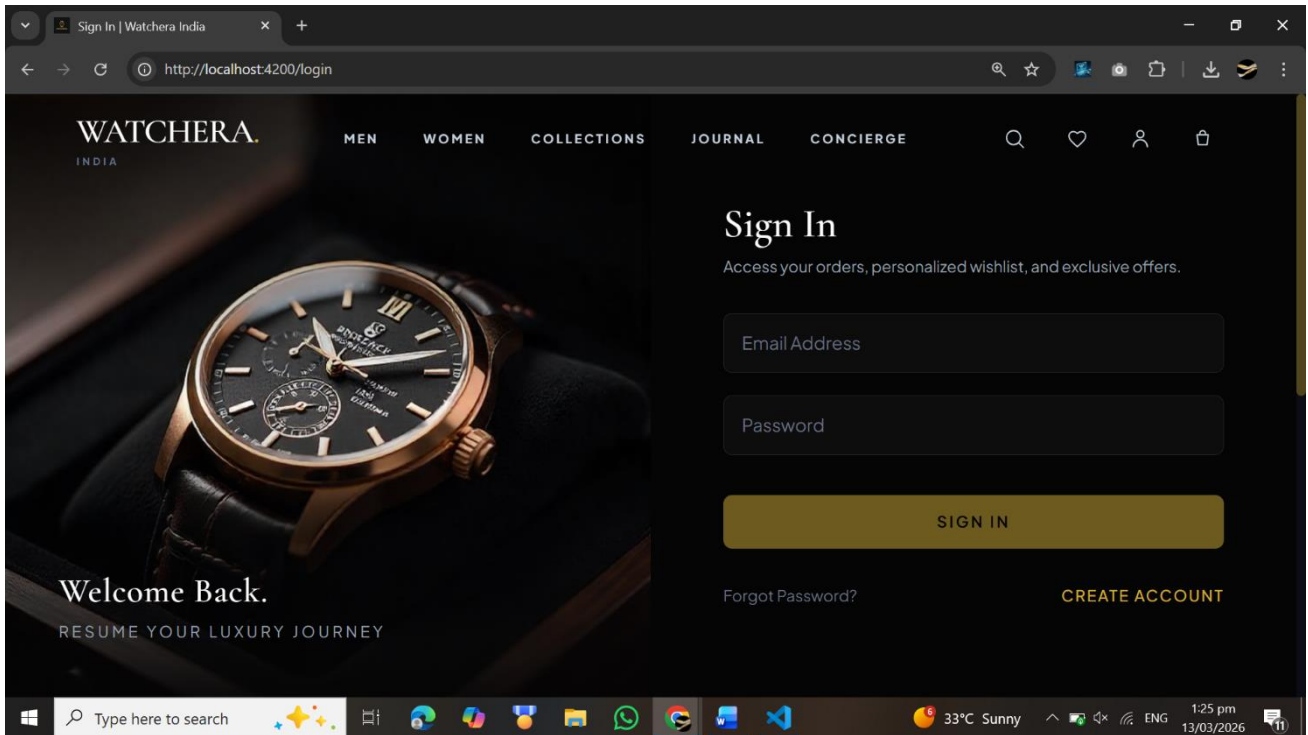
VIEW ALL WATCHES →

Men

Women

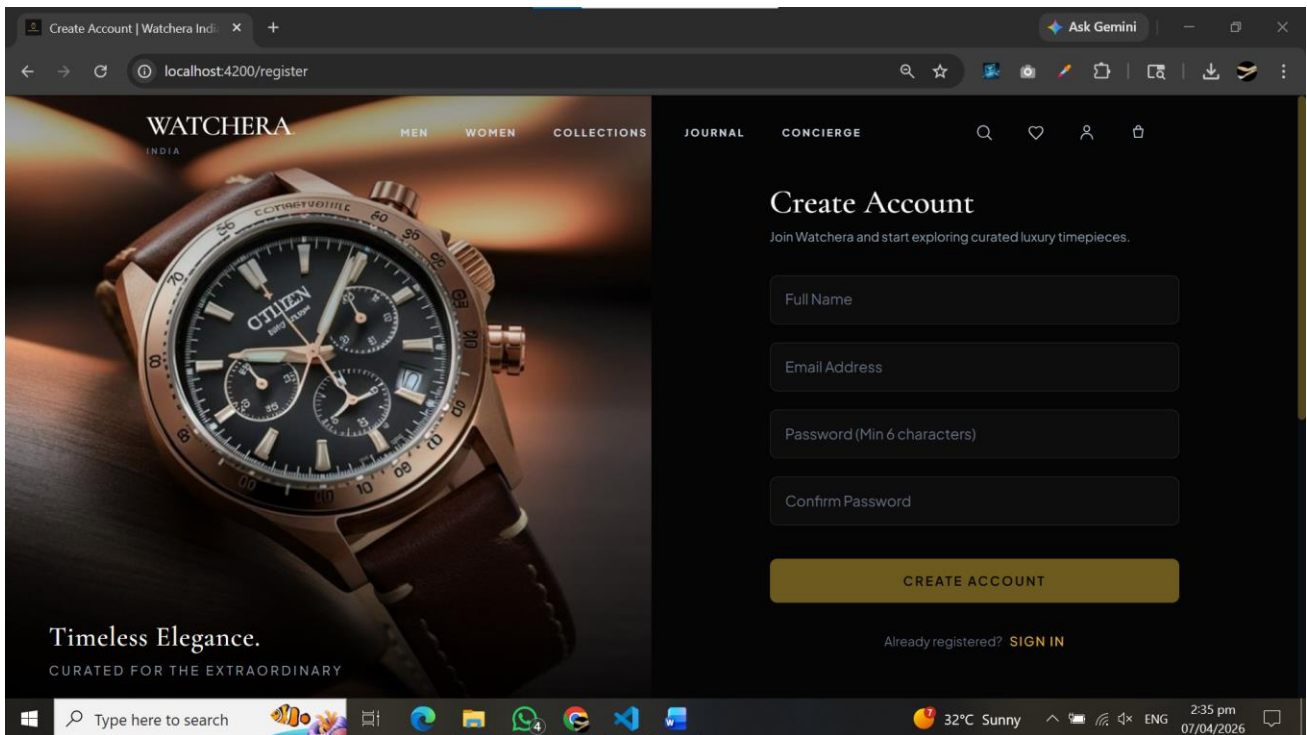
All Collections

Login page :



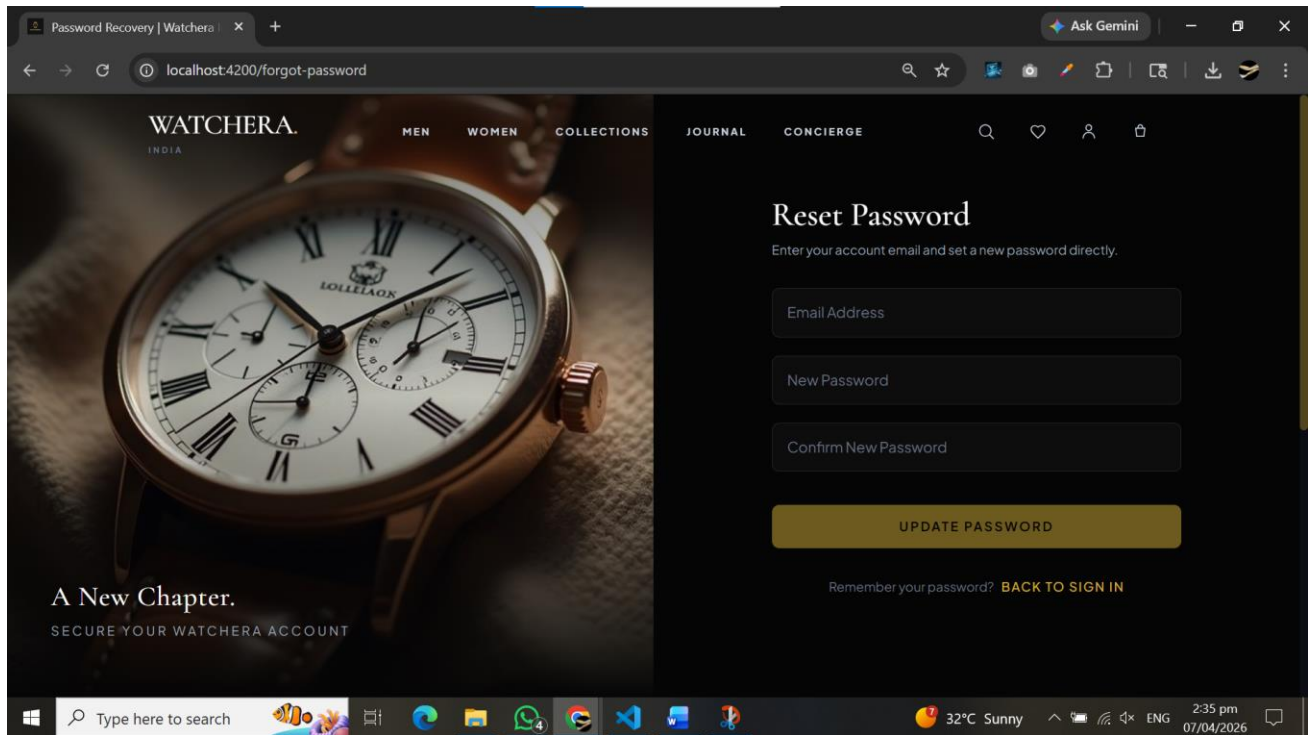
The screenshot shows the 'Sign In' page of the Watchera India website. The browser address bar displays 'http://localhost:4200/login'. The page features a dark theme with a large image of a luxury watch on the left. The navigation menu includes 'MEN', 'WOMEN', 'COLLECTIONS', 'JOURNAL', and 'CONCIERGE'. The 'Sign In' section includes a subtitle 'Access your orders, personalized wishlist, and exclusive offers.', followed by input fields for 'Email Address' and 'Password', a 'SIGN IN' button, and links for 'Forgot Password?' and 'CREATE ACCOUNT'. The footer shows the Windows taskbar with the date 13/03/2026 and time 1:25 pm.

Registration page :

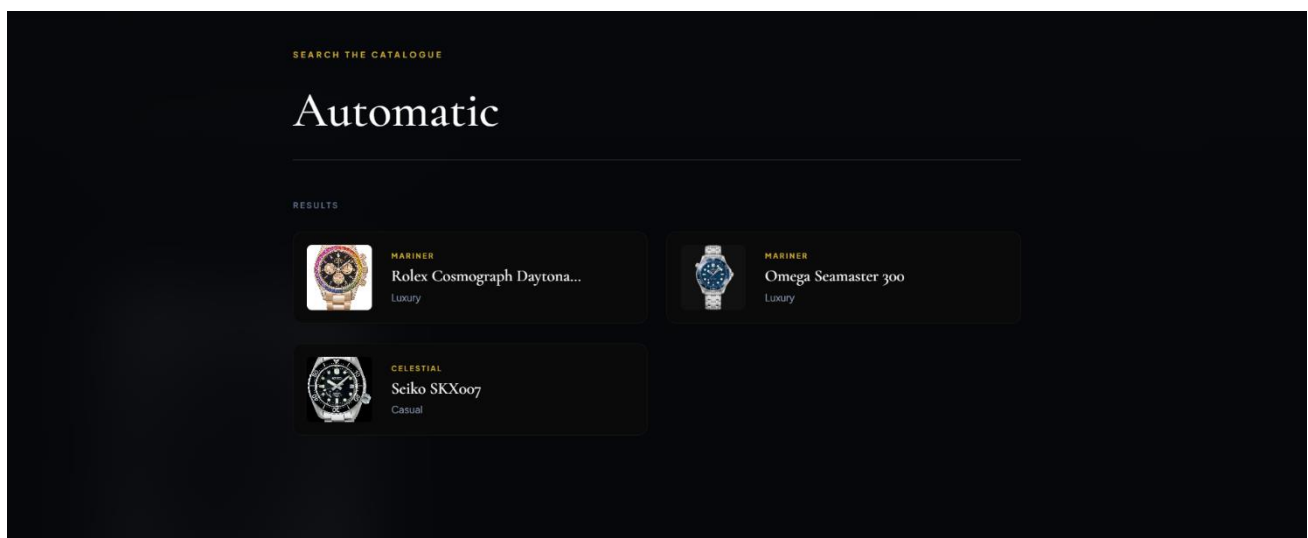


The screenshot shows the 'Create Account' page of the Watchera India website. The browser address bar displays 'localhost:4200/register'. The page features a dark theme with a large image of a luxury watch on the left. The navigation menu includes 'MEN', 'WOMEN', 'COLLECTIONS', 'JOURNAL', and 'CONCIERGE'. The 'Create Account' section includes a subtitle 'Join Watchera and start exploring curated luxury timepieces.', followed by input fields for 'Full Name', 'Email Address', 'Password (Min 6 characters)', and 'Confirm Password', a 'CREATE ACCOUNT' button, and a link for 'Already registered? SIGN IN'. The footer shows the Windows taskbar with the date 07/04/2026 and time 2:35 pm.

Forgot Password page :



Search feature :



Collection page :

WATCHERA.
INDIA

[MEN](#)
[WOMEN](#)
[COLLECTIONS](#)
[JOURNAL](#)
[CONCIERGE](#)

CURATED CATALOGUE

Timepieces

Filter by brand, price, material, and movement to discover the perfect addition to your collection.

SHOWING 7 RESULTS

Refine Search

BRANDS

☐ Diesel
 ☐ Omega
 ☐ Rolex
 ☐ Seiko
 ☐ Seiko
 ☐ Watchera

PRICE

☐ Under ₹50,000
 ☐ ₹50,000 - ₹1,00,000
 ☐ ₹1,00,000 - ₹2,00,000
 ☐ Above ₹2,00,000

COLLECTION

☐ Palmetto



SEIKO

Skondra Velocity Men's Chronograph Two-Tone Stainless Steel Watch

Japanese Quartz Chronog...

₹24,500



DIESEL

Mega Chief Analog Chronograph Men's Watch - Gold/Black

Quartz Chronograph - Sta...

₹22,000



ROLEX

Cosmograph Daytona "Rainbow"

Rolex Caliber 4130 Self-w...

₹100,000



WATCHERA

Serpentina Royale Gold

Swiss Quartz Movement - 1...

₹249,999



WATCHERA

Celsia Diamond Rose Gold

Swiss Quartz Movement - 1...

₹29,999



OMEGA

Seamaster 300 Automatic

Automatic - Stainless Steel

₹50,000



SEIKO

SKX007 Automatic

Automatic - Stainless Steel

₹35,000

WATCHERA.

Modern Indian watchmaking with global finishing standards. Every piece is authenticated and supported by the Watchera concierge.

SHOP

[All Collections](#)
[Men's Watches](#)
[Women's Watches](#)
[The Journal](#)

SUPPORT

[FAQ](#)
[Track Order](#)
[Shipping & Returns](#)
[Warranty](#)

INNER CIRCLE

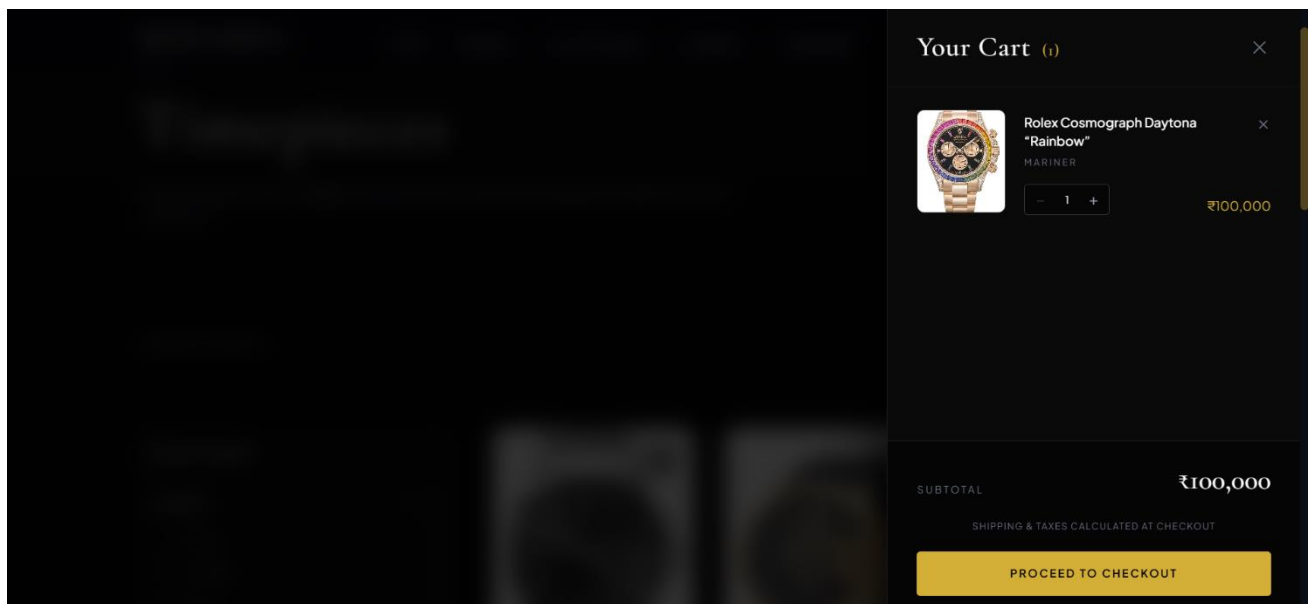
Get launch previews, exclusive event invites, and personalized concierge offers in India.

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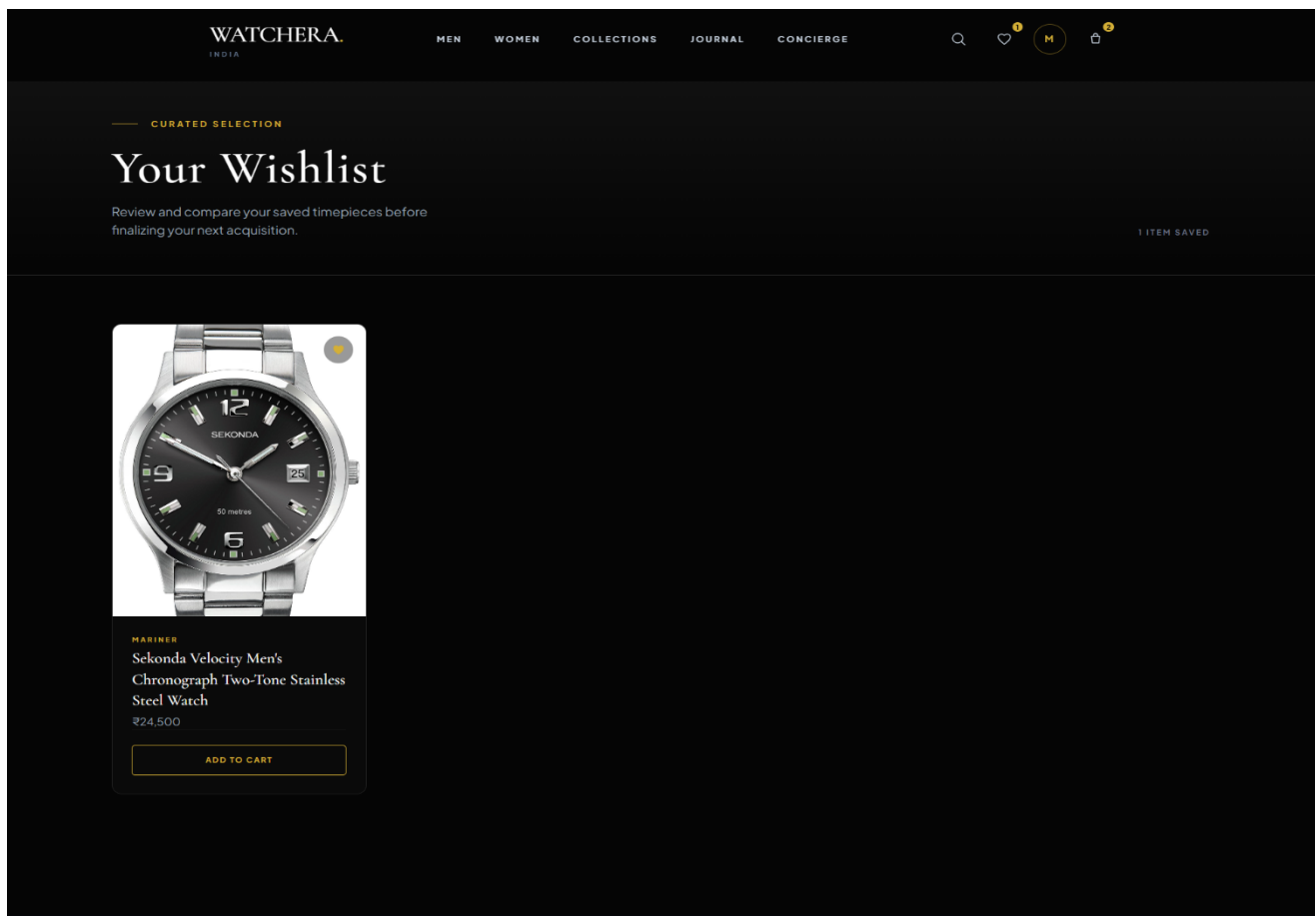
[Privacy Policy](#)
[Terms of Service](#)

35

Cart page :



Wishlist page :



Product Info page :

WATCHERA

INDIA

MEN

WOMEN

COLLECTIONS

JOURNAL

CONCIERGE

Q

♡

M

🏠

HOME / COLLECTIONS / ROLEX COSMOGRAPH DAYTONA "RAIN...





MARINER

Rolex Cosmograph Daytona
"Rainbow"

SKU: 69553JA

₹100,000

INCL. OF ALL TAXES

100% Genuine Products

Secure Payments

The Rolex Cosmograph Daytona Rainbow is one of the most exclusive and visually striking timepieces ever created by Rolex. Crafted in 18 ct Everose Gold, this model features a spectacular bezel set with 36 baguette-cut sapphires arranged in a perfect rainbow gradient. The lugs are adorned with brilliant-cut diamonds, elevating its luxurious presence. The black dial is accented with rainbow sapphire hour markers, complementing the vibrant bezel while maintaining sporty chronograph functionality. Powered by Rolex's in-house Caliber 4130 automatic chronograph movement, it offers exceptional precision, durability, and a 72-hour power reserve. This is not just a watch — it is a high-investment collector's piece.

ADD TO CART

WISHLIST

TIMEPIECE ESSENCE

CASE
18 ct Everose Gold (Rolex's proprietary rose gold alloy)

MOVEMENT
Rolex Caliber 4130 Self-winding mechanical chronograph movement

DIAL
Rainbow-colored baguette-cut sapphires (36 stones) Diamond set lugs

STRAP
18 ct Everose Gold Oyster Bracelet

DETAILED SPECIFICATIONS

CUSTOMER REVIEWS (1)

Technical Blueprint

Case Material	18 ct Everose Gold (Rolex's proprietary rose gold alloy)
Movement Caliber	Rolex Caliber 4130 Self-winding mechanical chronograph movement
Dial Dimensions	40mm
Water Resistance Rating	100 meters (10 ATM)
Manufacturer Warranty	2 Years

Distinguished Features

- Rolex Caliber 4130 Self-winding mechanical chronograph movement
- 18 ct Everose Gold (Rolex's proprietary rose gold alloy) case
- 18 ct Everose Gold Oyster Bracelet strap
- Water resistance 100 meters (10 ATM)

DISCOVER MORE CURATED TIMEPIECES

Related Watches



MARINER

Sekonda Velocity Men's Chronograph Two-Tone Stainless Steel Watch

₹24,500



MARINER

Diesel Mega Chief Analog Chronograph Men's Watch - Gold/Black

₹22,000



MARINER

Watchera Serpentina Royale Gold

₹29,999

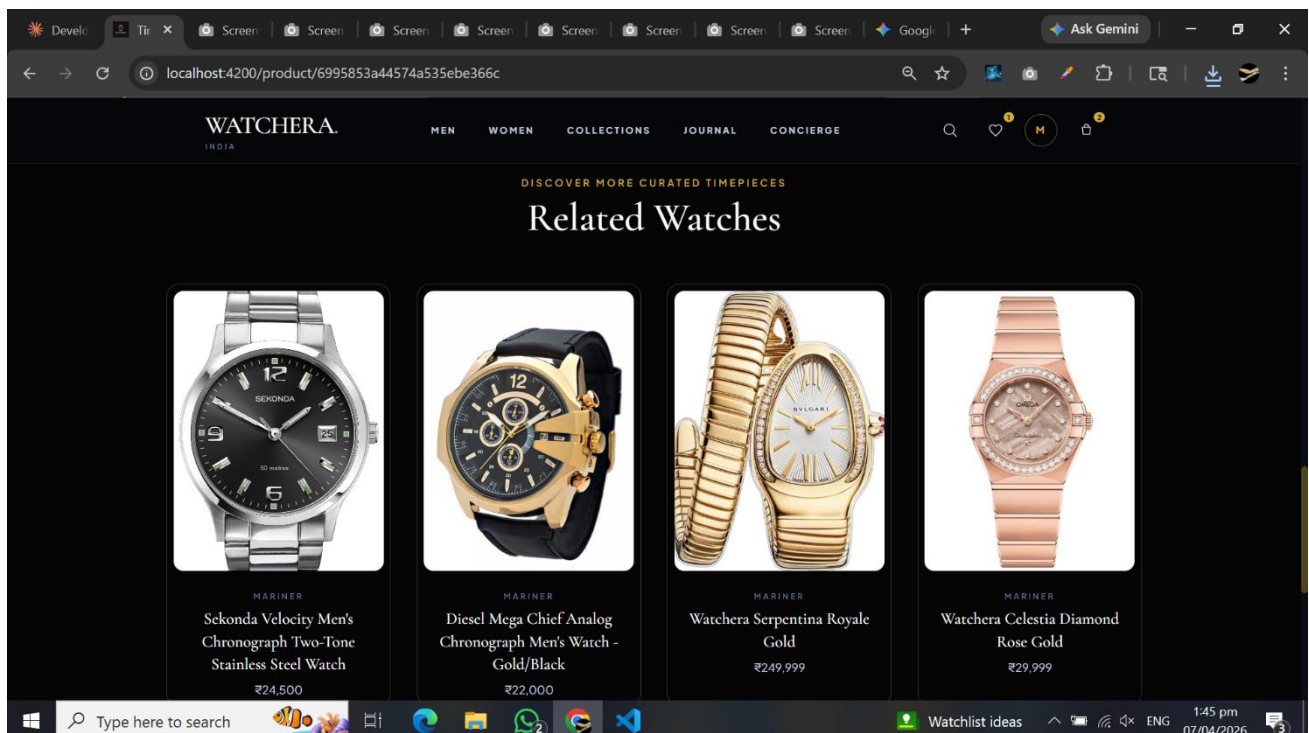
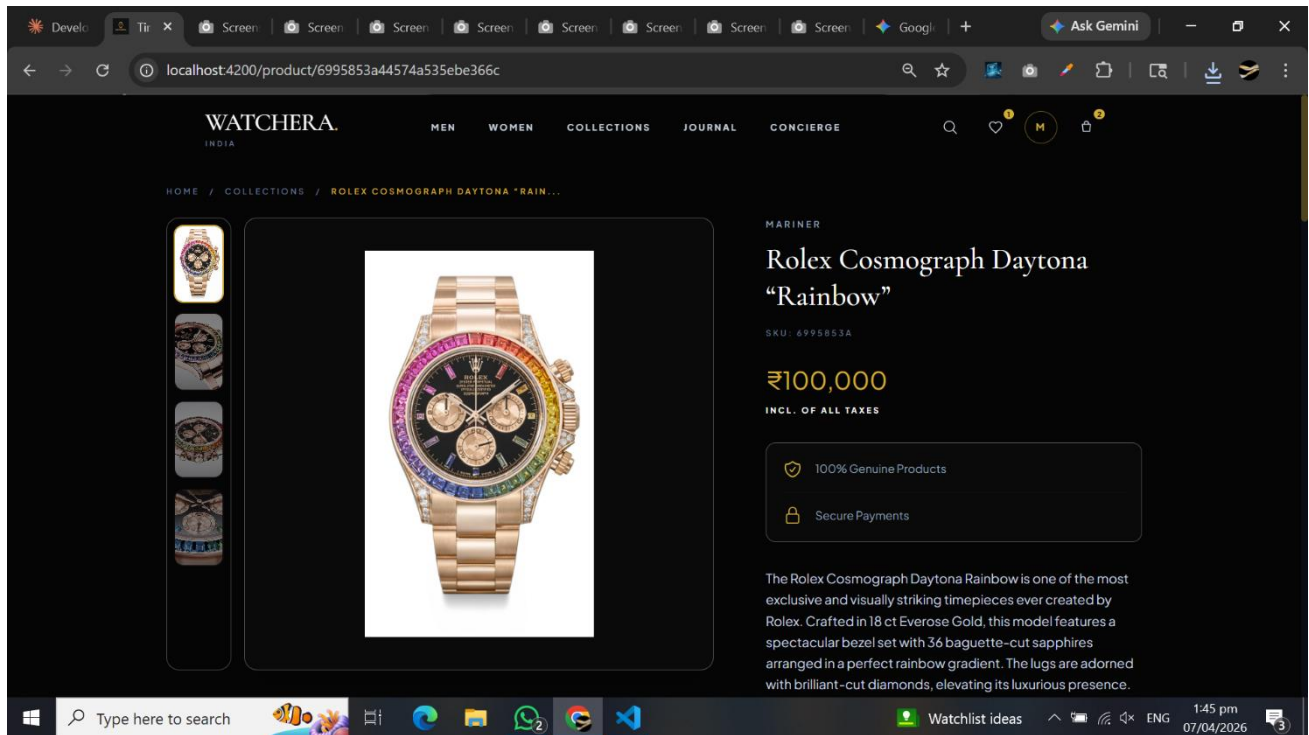


MARINER

Watchera Celestia Diamond Rose Gold

₹29,999

Product Info :



User Dashboard :

WATCHERA
INDIA

MENWOMENCOLLECTIONSJOURNALCONCIERGE

CLIENT PORTAL

My Dashboard

Manage your account details, curate your wishlist, and track your recent watch acquisitions in one secure place.

TOTAL ORDERS
2

TOTAL SPENT
₹248,999

WISHLIST
1

LAST ORDER
Mar 25, 2026

Manthan Patel
manthan@gmail.com

RECENT ORDERS

CLIENT PROFILE
WISHLIST
1
SIGN OUT

Recent Orders

Review your latest purchases and their delivery status.

ORDER PLACED
Mar 25, 2026

ORDER NUMBER
#7c3499

TOTAL AMOUNT
₹33,000

VIEW RECEIPT →

ORDER PLACED
Mar 14, 2026

ORDER NUMBER
#df4685

TOTAL AMOUNT
₹215,999

VIEW RECEIPT →

Require Assistance?

Contact our concierge team for order modifications, tracking details, or personalized recommendations.

CONTACT US

Order History :

WATCHERA.
INDIA

MEN WOMEN COLLECTIONS JOURNAL CONCIERGE






CLIENT ACCOUNT

Order History

Track your recent Watchera acquisitions and delivery status.

CONTINUE SHOPPING

ORDER DATE	ORDER ID	TOTAL	
Mar 25, 2026	#7c3499	₹33,000	CONFIRMED
 QTY: 1 Seiko SKX007 ₹33,000			
Mar 14, 2026	#df4685	₹239,999	SHIPPED
 QTY: 1 Watchera Serpentina Royale Gold ₹239,999			

WATCHERA.
Modern Indian watchmaking with global finishing standards. Every piece is authenticated and supported by the Watchera concierge.

SHOP

All Collections
Men's Watches
Women's Watches
The Journal

SUPPORT

FAQ
Track Order
Shipping & Returns
Warranty

INNER CIRCLE

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Checkout Page :

WATCHERA
INDIA

MENWOMENCOLLECTIONSJOURNALCONCIERGE

M

WATCHERA

Secure Checkout

Complete your shipping details to finalize your Watchera order.

Shipping Destination

FULL NAME

Manthan Patel

PHONE NUMBER

+91 6354843201

POSTAL CODE

395004

DETAILED ADDRESS

A/2 Madhav Mandir Soc., Near Govindji Hall / SMVS BRTS, Dabholi - 395004, Surat, Gujarat

CITY

Surat

STATE

Gujarat

COUNTRY

India

PAY SECURELY WITH RAZORPAY

256-BIT SSL ENCRYPTED CHECKOUT

Order Summary



Omega Seamaster 300
QTY: 2

₹90,000

GIFT CARD / PROMO CODE

WATCH100

REMOVE

APPLY

Code applied! You save ₹13,500.

Subtotal

₹90,000

Shipping

FREE

Discount

-₹13,500

Total

₹76,500

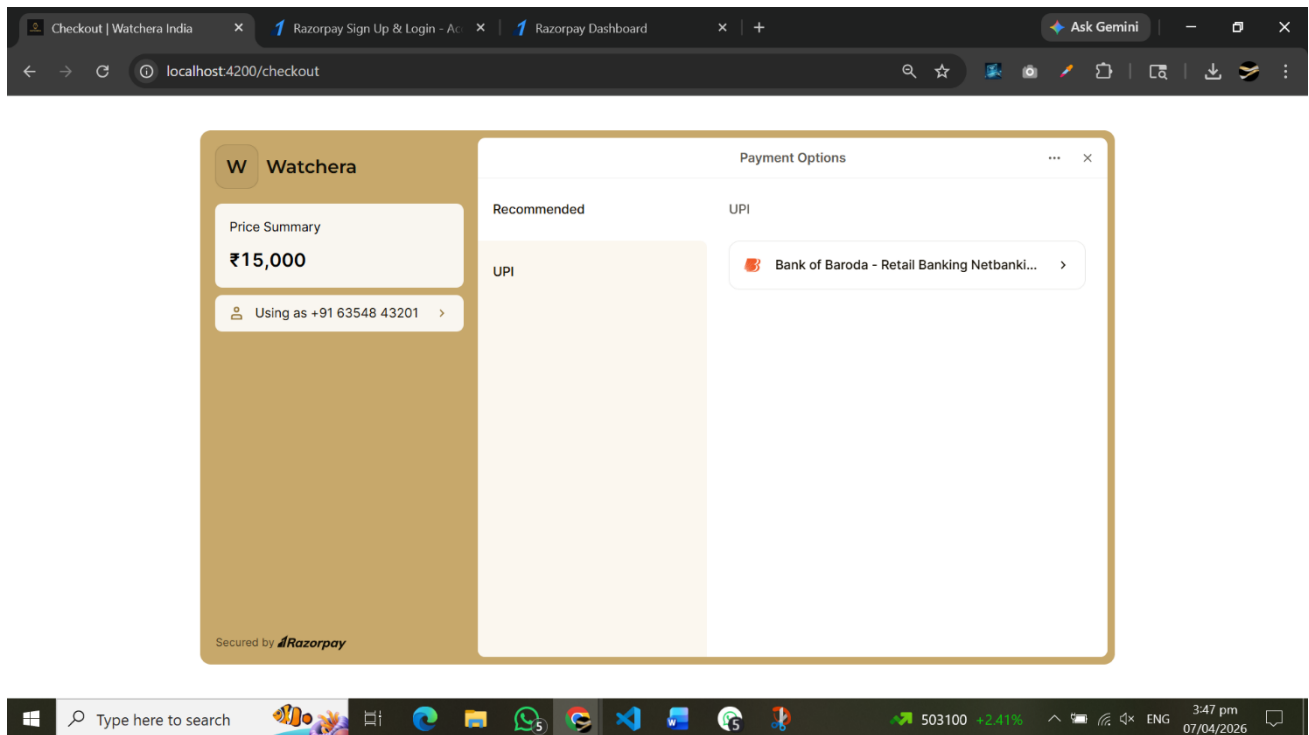
Advance to Pay Now

₹15,000

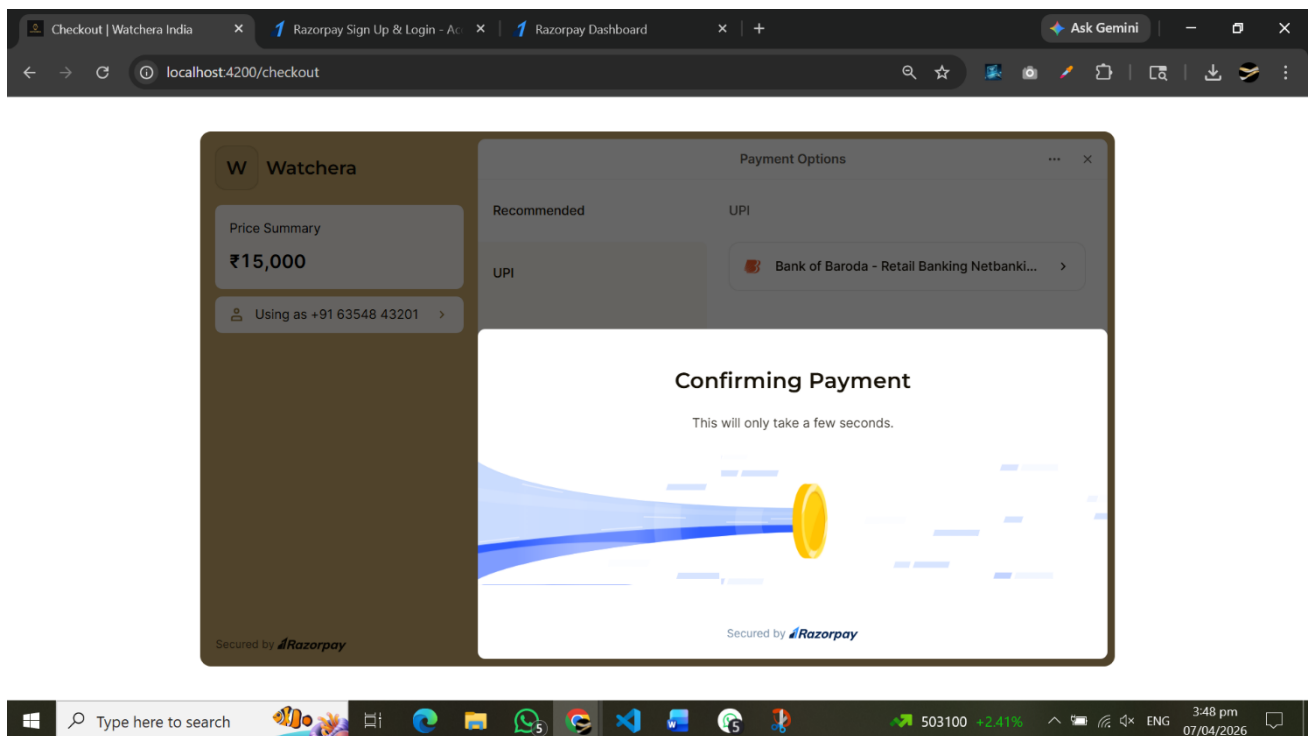
CASH ON DELIVERY

₹61,500

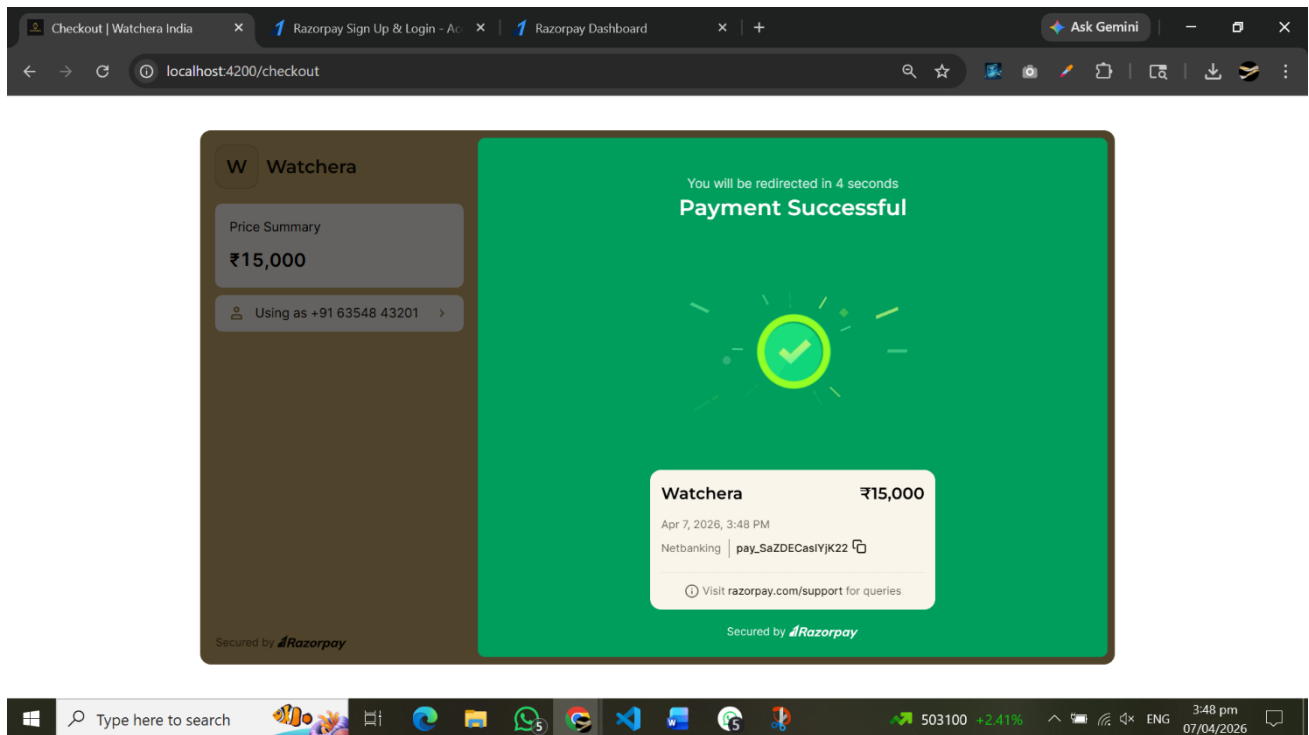
Razorpay Payment :



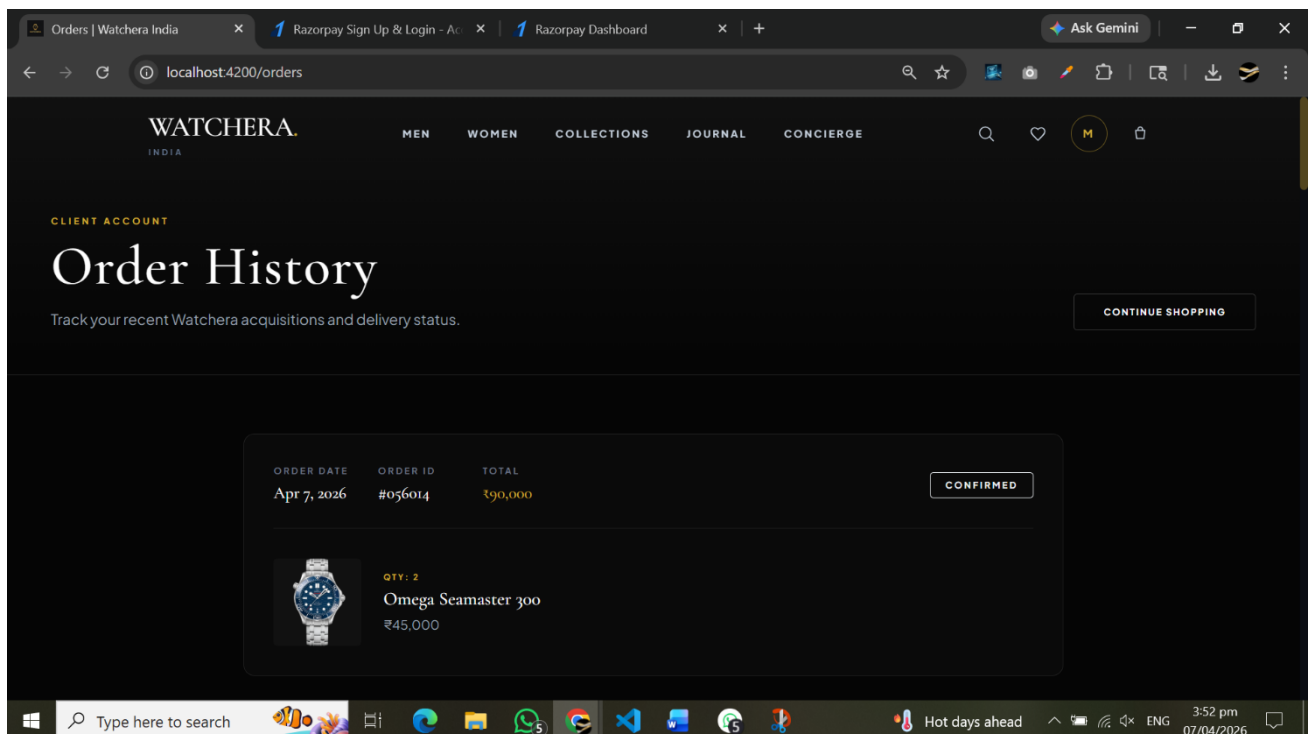
Processing Payments :



Success Payment :

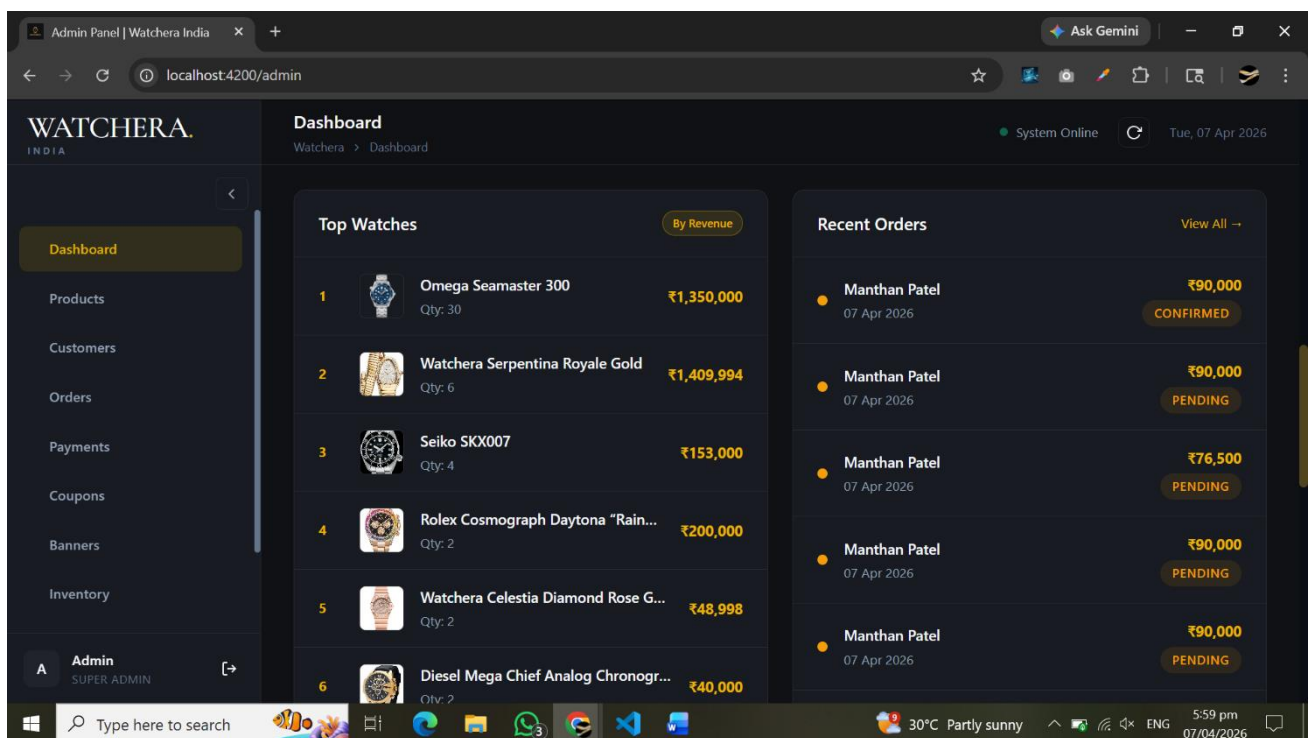
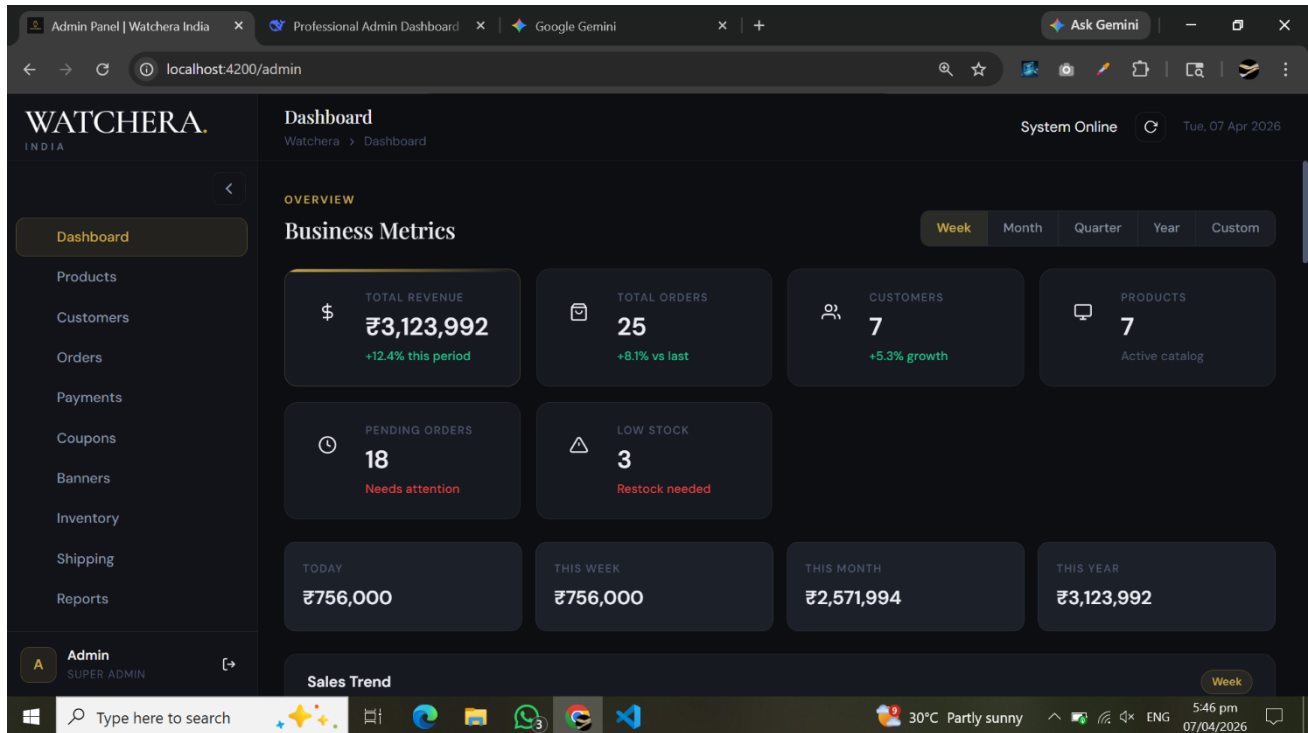


Order History :

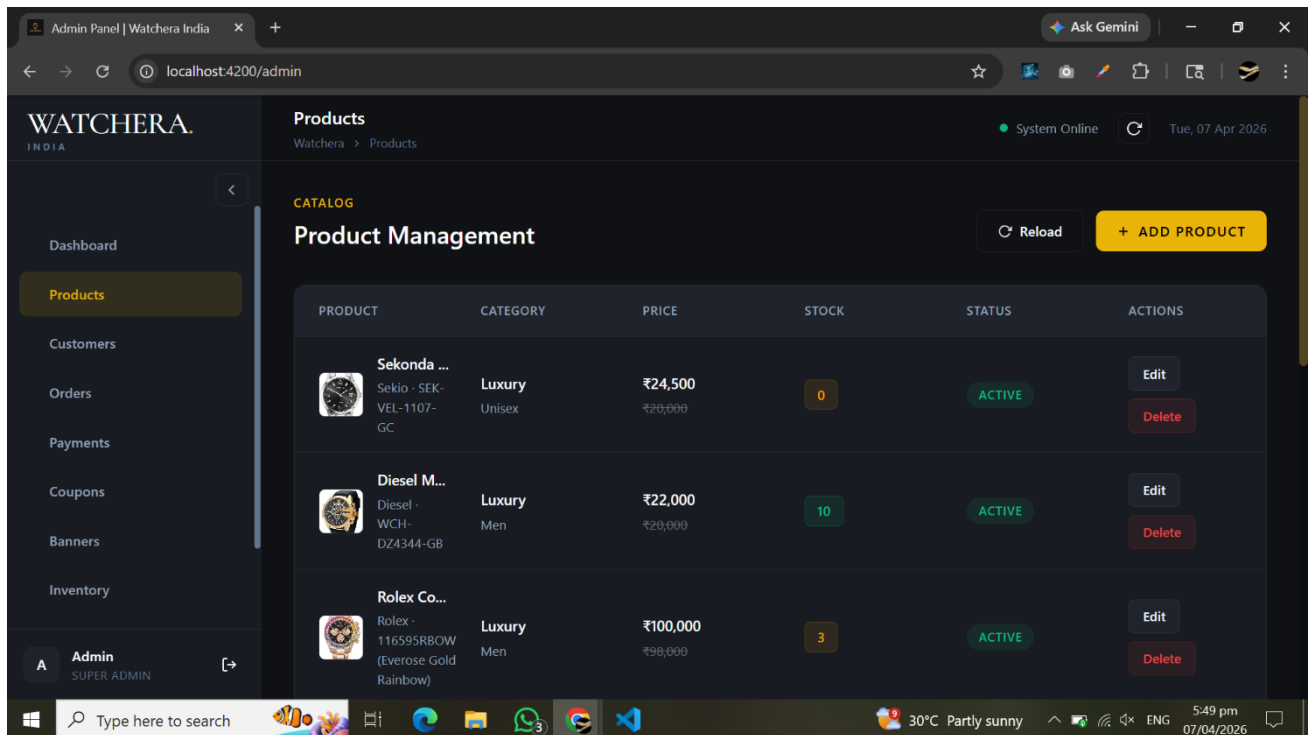


Admin Panel

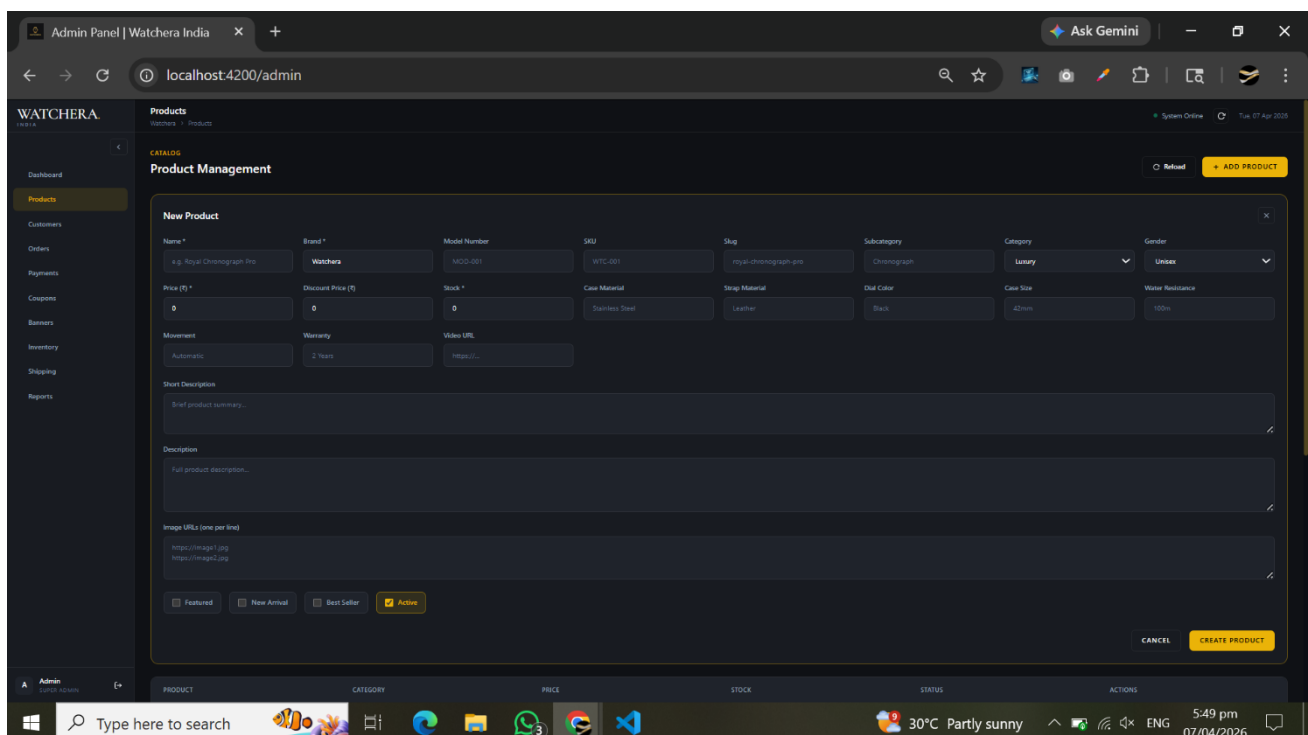
Admin Dashboard :



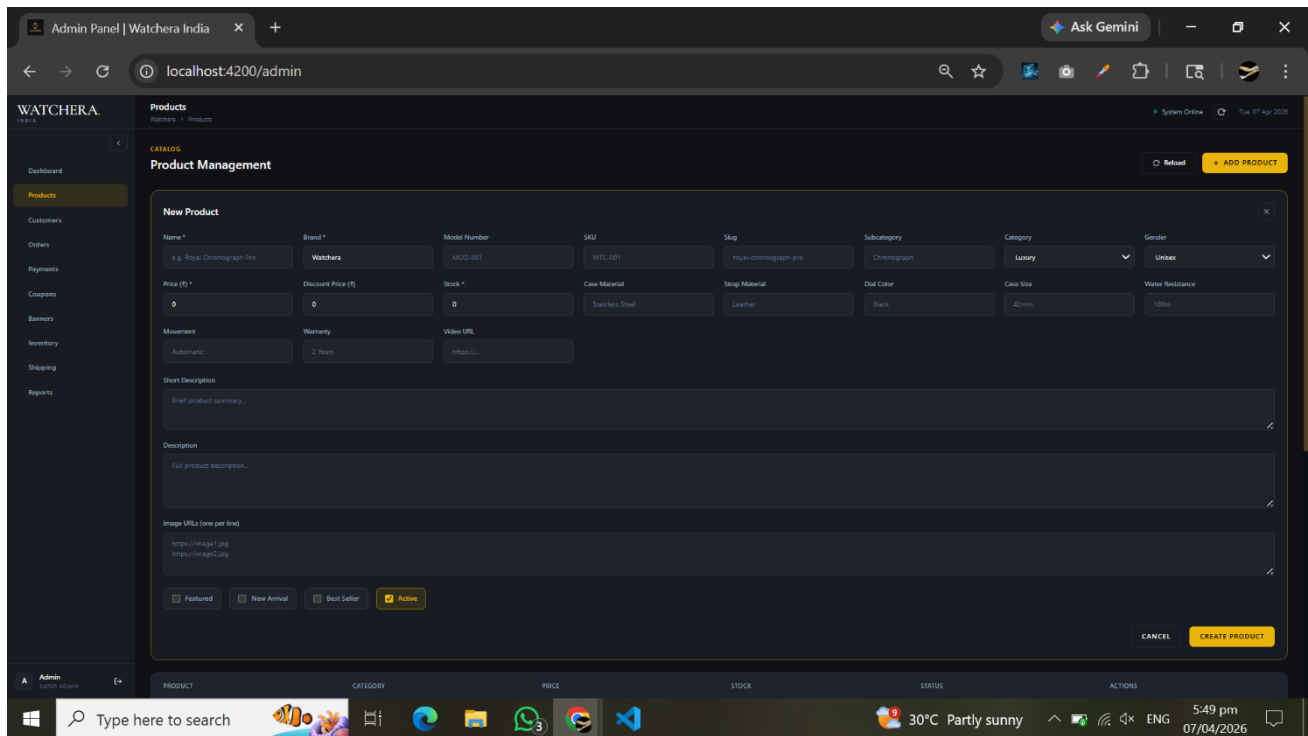
Admin Product Layout :



Admin Product ADD Layout :



Admin Customer Layout :



WATCHERA INDIA

Admin Panel | Watchera India

localhost:4200/admin

System Online Tue, 07 Apr 2026

Products

Product Management

New Product

Name * e.g. Royal Chronograph Pro Brand * Watchera Model Number MCG-001 SKU WTC-001 Shop royal-chronograph-pro Subcategory Chronograph Category Luxury Gender Unisex

Price (₹) * 0 Discount Price (%) 0 Stock * 0 Case Material Stainless Steel Strap Material Leather Dial Color Black Case Size 42mm Water Resistance 100m

Movement Automatic Warranty 2 Years Video URL https://

Short Description Brief product summary...

Description Full product description...

Image URLs (one per line) https://image1.jpg https://image2.jpg

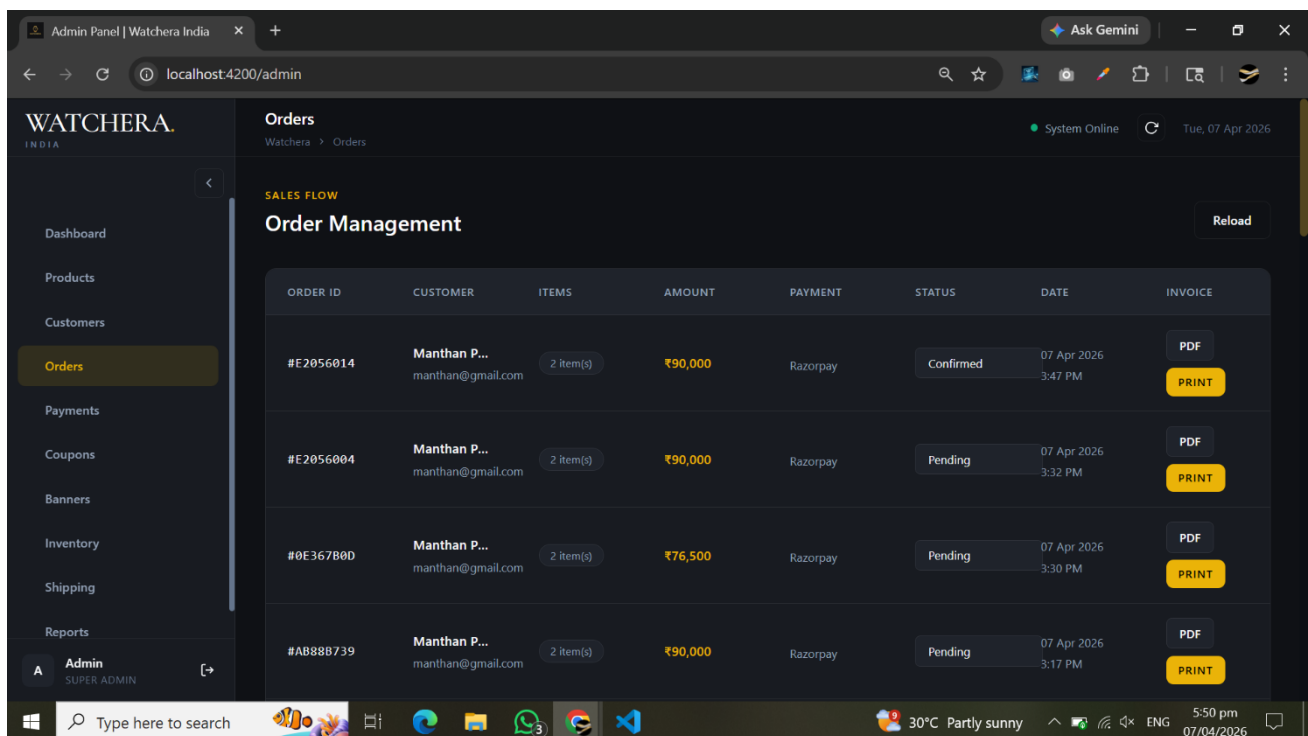
Featured New Arrival Best Seller Active

CANCEL CREATE PRODUCT

PRODUCT CATEGORY PRICE STOCK STATUS ACTIONS

Type here to search 30°C Partly sunny 5:49 pm 07/04/2026

Admin Order Layout :



WATCHERA INDIA

Admin Panel | Watchera India

localhost:4200/admin

System Online Tue, 07 Apr 2026

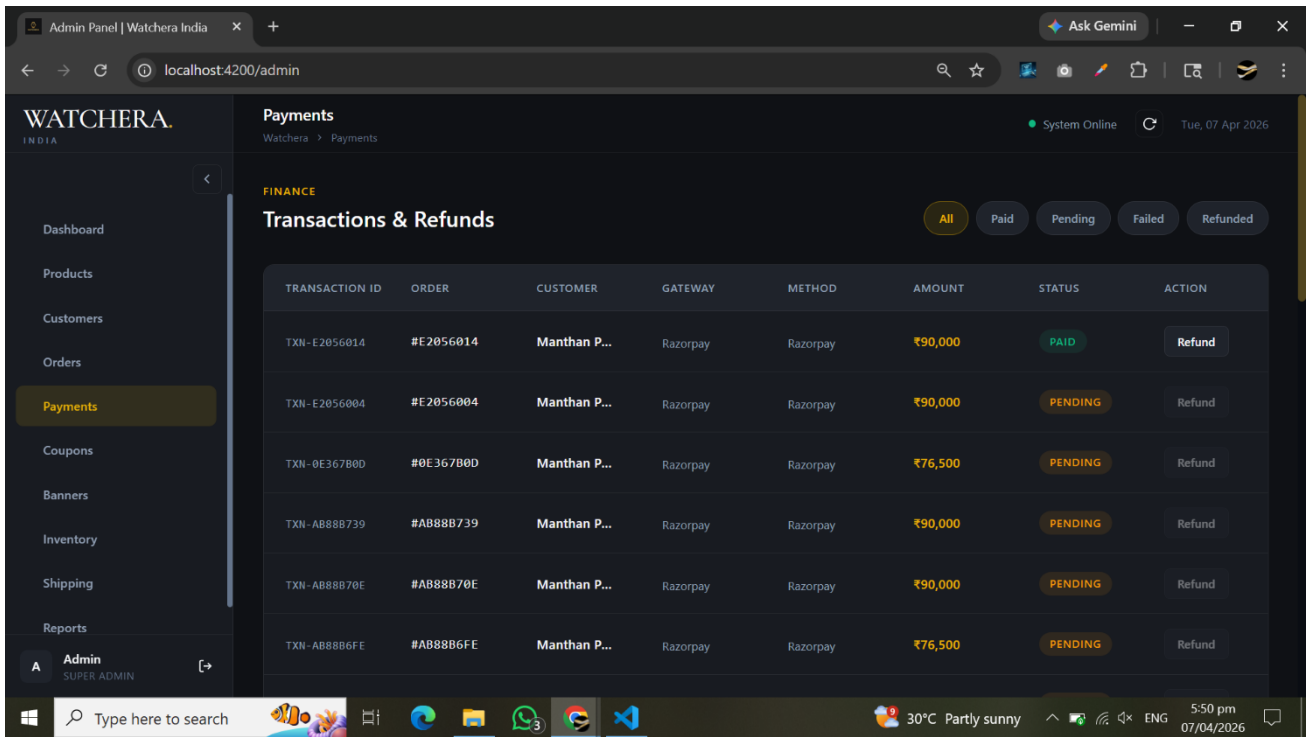
Orders

Order Management

ORDER ID	CUSTOMER	ITEMS	AMOUNT	PAYMENT	STATUS	DATE	INVOICE
#E2056014	Manthan P... manthan@gmail.com	2 Item(s)	₹90,000	Razorpay	Confirmed	07 Apr 2026 3:47 PM	PDF PRINT
#E2056004	Manthan P... manthan@gmail.com	2 Item(s)	₹90,000	Razorpay	Pending	07 Apr 2026 3:32 PM	PDF PRINT
#0E36780D	Manthan P... manthan@gmail.com	2 Item(s)	₹76,500	Razorpay	Pending	07 Apr 2026 3:30 PM	PDF PRINT
#AB888739	Manthan P... manthan@gmail.com	2 Item(s)	₹90,000	Razorpay	Pending	07 Apr 2026 3:17 PM	PDF PRINT

Type here to search 30°C Partly sunny 5:50 pm 07/04/2026

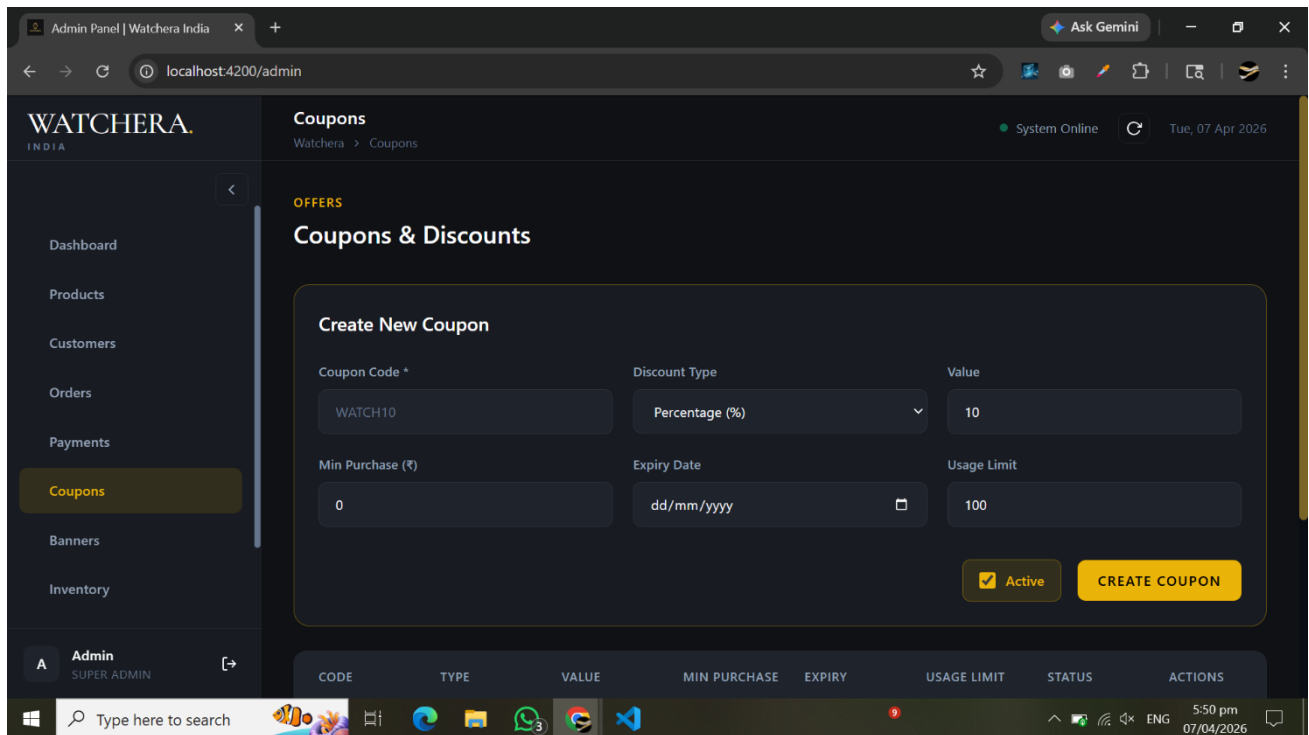
Admin Payment Layout :



The screenshot displays the 'Payments' section of the Watchera Admin Panel. The interface includes a sidebar with navigation options like Dashboard, Products, Customers, Orders, Payments (selected), Coupons, Banners, Inventory, Shipping, and Reports. The main content area shows a table of transactions and refunds, with filters for All, Paid, Pending, Failed, and Refunded. The table lists transaction details including Transaction ID, Order ID, Customer Name, Gateway, Method, Amount, Status, and an Action button for Refund.

TRANSACTION ID	ORDER	CUSTOMER	GATEWAY	METHOD	AMOUNT	STATUS	ACTION
TXN-E2056014	#E2056014	Manthan P...	Razorpay	Razorpay	₹90,000	PAID	Refund
TXN-E2056004	#E2056004	Manthan P...	Razorpay	Razorpay	₹90,000	PENDING	Refund
TXN-0E367B0D	#0E367B0D	Manthan P...	Razorpay	Razorpay	₹76,500	PENDING	Refund
TXN-AB88B739	#AB88B739	Manthan P...	Razorpay	Razorpay	₹90,000	PENDING	Refund
TXN-AB88B70E	#AB88B70E	Manthan P...	Razorpay	Razorpay	₹90,000	PENDING	Refund
TXN-AB88B6FE	#AB88B6FE	Manthan P...	Razorpay	Razorpay	₹76,500	PENDING	Refund

Admin Coupons Layout :



Admin Panel | Watchera India

localhost:4200/admin

WATCHERA INDIA

Coupons

System Online Tue, 07 Apr 2026

OFFERS

Coupons & Discounts

Create New Coupon

Coupon Code * WATCH10

Discount Type Percentage (%)

Value 10

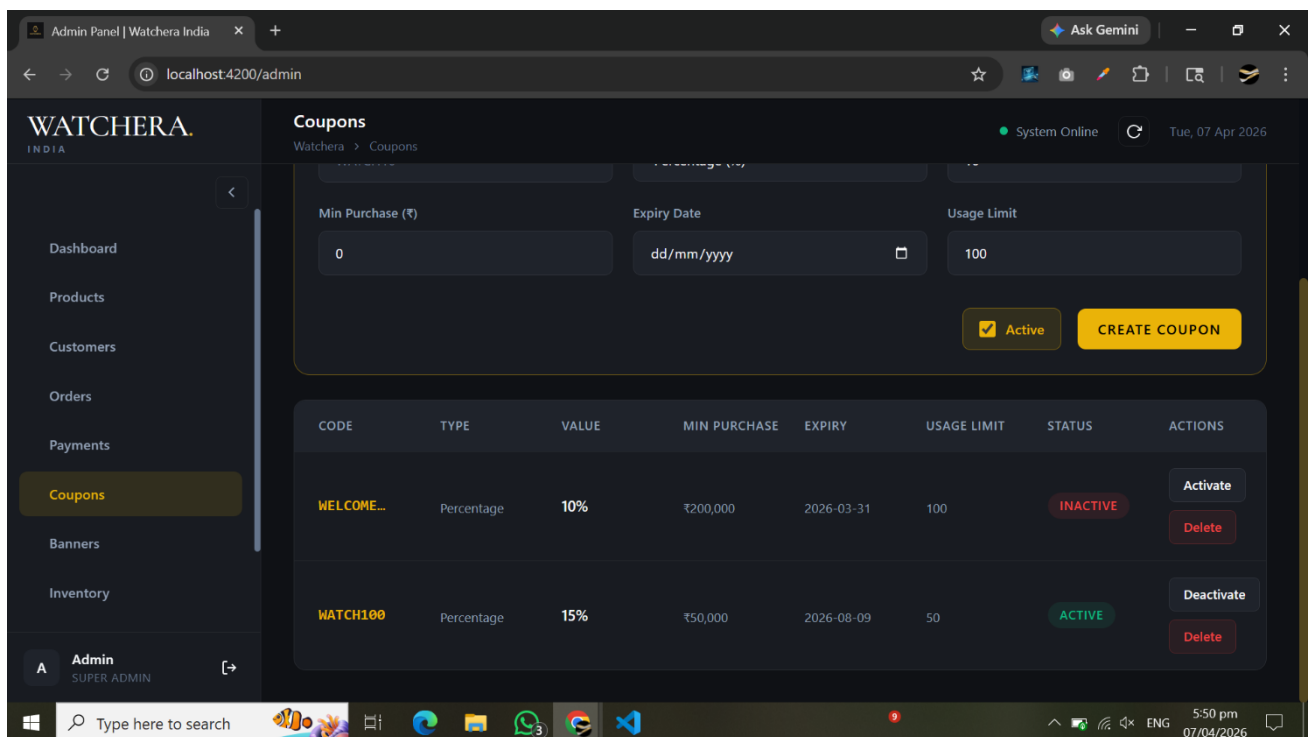
Min Purchase (₹) 0

Expiry Date dd/mm/yyyy

Usage Limit 100

☒ Active **CREATE COUPON**

CODE	TYPE	VALUE	MIN PURCHASE	EXPIRY	USAGE LIMIT	STATUS	ACTIONS
------	------	-------	--------------	--------	-------------	--------	---------



Admin Panel | Watchera India

localhost:4200/admin

WATCHERA INDIA

Coupons

System Online Tue, 07 Apr 2026

OFFERS

Coupons & Discounts

Create New Coupon

Coupon Code * WATCH10

Discount Type Percentage (%)

Value 10

Min Purchase (₹) 0

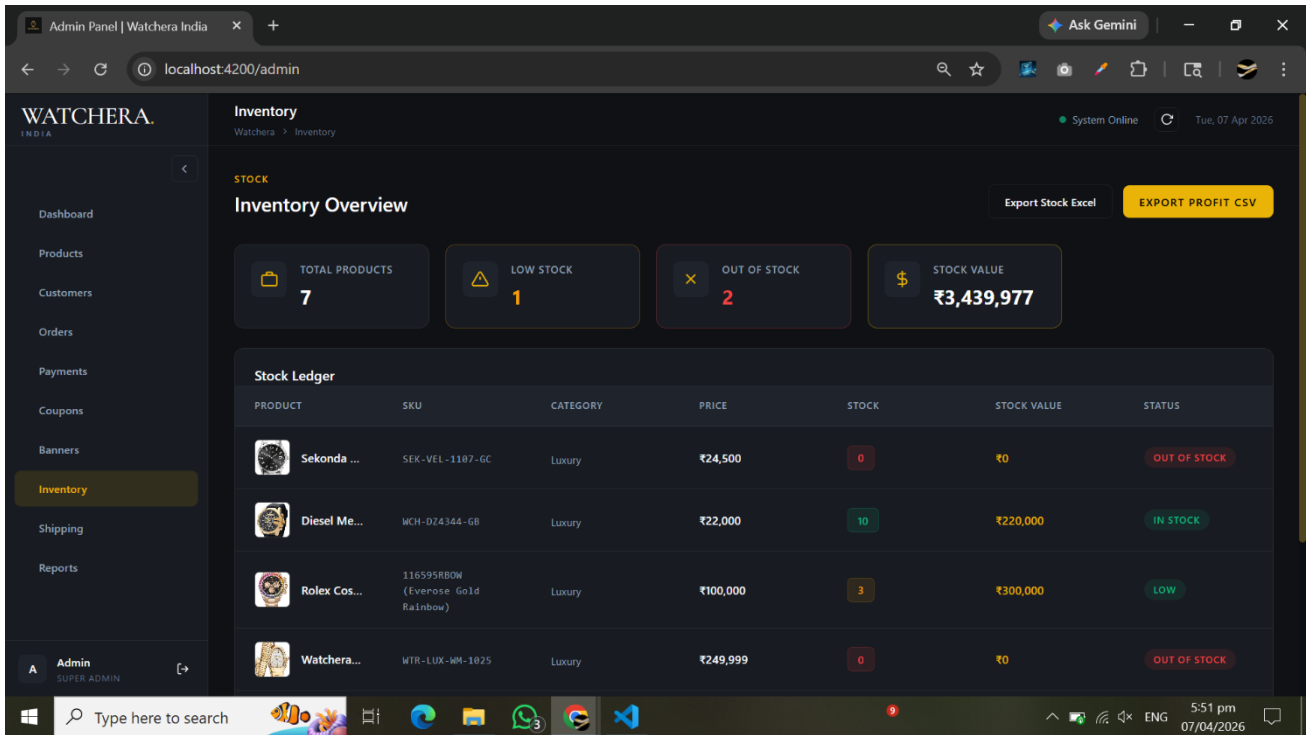
Expiry Date dd/mm/yyyy

Usage Limit 100

☒ Active **CREATE COUPON**

CODE	TYPE	VALUE	MIN PURCHASE	EXPIRY	USAGE LIMIT	STATUS	ACTIONS
WELCOME...	Percentage	10%	₹200,000	2026-03-31	100	INACTIVE	Activate Delete
WATCH100	Percentage	15%	₹50,000	2026-08-09	50	ACTIVE	Deactivate Delete

Admin Inventory Layout :

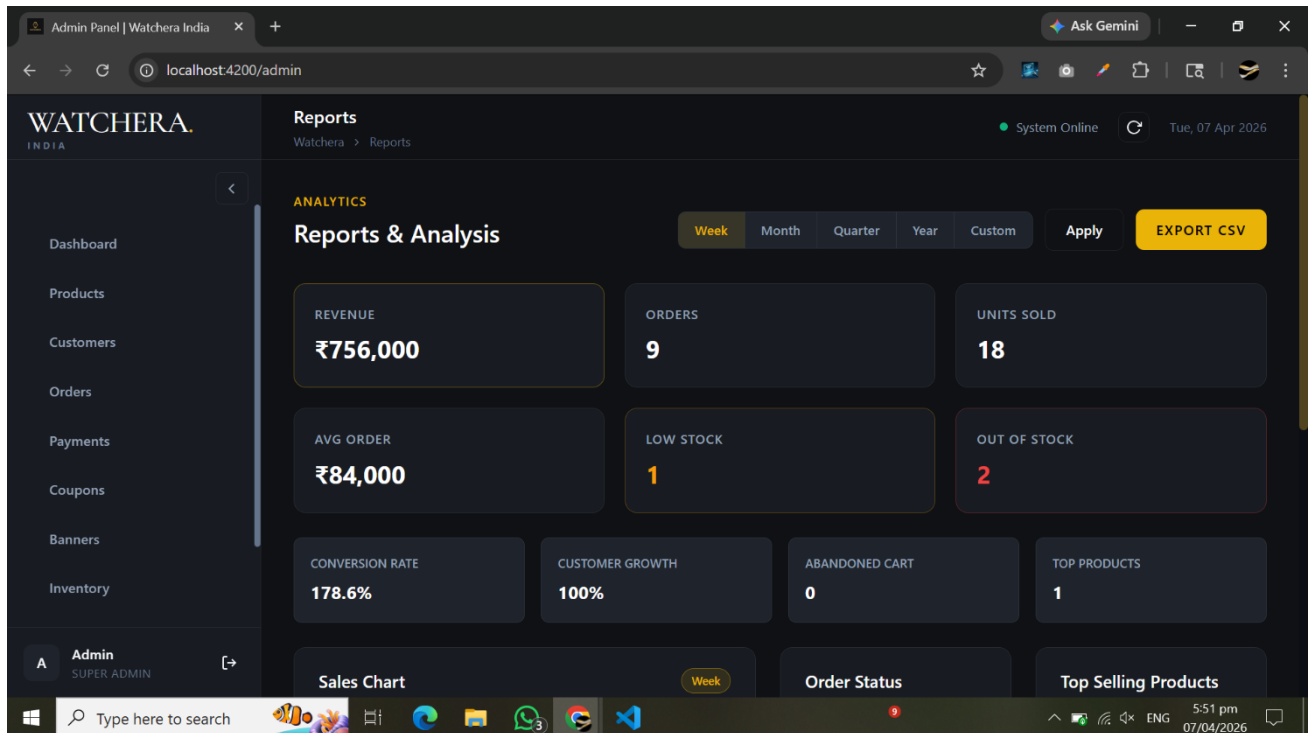


The screenshot displays the 'Inventory' section of the Watchera Admin Panel. The interface includes a sidebar with navigation options: Dashboard, Products, Customers, Orders, Payments, Coupons, Banners, Inventory (highlighted), Shipping, and Reports. The main content area shows the 'Inventory Overview' with four summary cards: TOTAL PRODUCTS (7), LOW STOCK (1), OUT OF STOCK (2), and STOCK VALUE (₹3,439,977). Below these is a 'Stock Ledger' table listing products, their SKUs, categories, prices, stock levels, stock values, and statuses.

PRODUCT	SKU	CATEGORY	PRICE	STOCK	STOCK VALUE	STATUS
Sekonda ...	SEK-VEL-1187-GC	Luxury	₹24,500	0	₹0	OUT OF STOCK
Diesel Me...	WCH-DZ4344-GB	Luxury	₹22,000	10	₹220,000	IN STOCK
Rolex Cos...	116595RBOW (Everose Gold Rainbow)	Luxury	₹100,000	3	₹300,000	LOW
Watchera...	WTR-LUX-WM-1025	Luxury	₹249,999	0	₹0	OUT OF STOCK

The bottom of the screen shows a Windows taskbar with a search bar, application icons, and system tray information including the time (5:51 pm) and date (07/04/2026).

Admin Report Layout :



Software Testing

Software testing is a critical phase in the development of the Watchera E-commerce Platform, ensuring that the system is robust, reliable, and meets all functional and non-functional requirements. The testing process was carried out systematically, focusing on both logical correctness and usability across different environments. The goal was to uncover potential errors, validate data integrity, and ensure that user interactions result in the expected behavior.

7.1 Objectives of Testing

- To verify that all web pages and features are functioning correctly.
- To ensure proper communication between the frontend web application, the backend API, and the MongoDB database.
- To confirm that the design and layout are responsive, maintaining a premium visual consistency across different devices and browsers.
- To detect and correct functional, logical, or performance-related issues before deployment.

7.2 Levels of Testing

- **Program Level Testing:** Each script and component was tested individually to confirm that it executes correctly and handles all expected inputs. For example, product addition, deletion, and coupon application in the admin panel were tested for proper database updates.
- **Module Level Testing:** All core modules of the website—such as Product Management, User Management, Order Processing, Cart & Wishlist Operations, and Coupon Management—were isolated and tested to ensure smooth interaction and correct data flow.
- **Integration Testing:** After module-level testing, integrated testing was performed to ensure that all modules interact correctly. The end-to-end functionality of the website, from adding a watch to the cart, applying a discount code, to final order placement and payment processing, was confirmed to work as intended.

7.3 Functional Testing :

Functional testing was performed to ensure that the website fulfills all specified requirements.

Key aspects tested include:

- Navigation menus, advanced category filters (e.g., brand, movement, gender), product detail pages, and checkout processes function as expected.
- Database operations for products, users, orders, and coupons perform accurately within the MongoDB environment.
- Shopping cart updates, wishlist modifications, order placement, and invoice generation are error-free.
- Admin dashboard reports display accurate, real-time data for Sales, Inventory Stock, Active Users, and Order Status.
- Responsive design validation on desktop, tablet, and mobile devices to ensure a seamless shopping experience.

7.4 Environment Testing The website was tested in different environments to ensure compatibility and stability:

- **Web Browsers:** Google Chrome, Safari, Microsoft Edge, Mozilla Firefox
- **Operating Systems:** Windows 11, macOS, iOS, Android
- **Backend Environment:** Node.js (Express)
- **Database:** MongoDB (Atlas)
- **Devices:** Desktop, Laptop, Tablet, and Mobile devices
- **Performance Testing:** Basic load testing was performed to ensure smooth operation, fast image loading for high-quality watch assets, and stability under normal traffic conditions.

7.5 Testing Outcomes

- No critical functional errors were detected during final staging.
- Minor UI/UX design adjustments were made to improve responsiveness and the premium feel of the mobile view.
- Database queries were fully verified and optimized for efficiency to ensure fast loading of product catalogs.
- User navigation and the end-to-end purchasing experience were tested and confirmed to be highly intuitive and seamless.

Future Scope of Enhancements

8.1 Future Scope of Enhancements

The Watchera platform is designed with scalability in mind and can be enhanced in multiple ways to improve usability, increase conversion rates, and elevate the luxury shopping experience:

1. **3D Visual Try-On Integration:** Implementing a real-time virtual try-on feature using React and 3D models. This will allow customers to use their device cameras to see how a specific luxury watch looks on their wrist before purchasing.
2. **Native Mobile Application:** Converting the robust web application into a dedicated Android application to provide a faster, more integrated mobile shopping experience and utilize native device features like push notifications.
3. **AI-Powered Product Imaging & Recommendations:** Integrating AI tools to dynamically upscale user-uploaded review images, automatically generate premium backgrounds for new watch listings, and provide smart, algorithm-driven watch recommendations based on browsing history.
4. **Expanded Payment & Security Ecosystem:** Supporting a wider array of global payment gateways, digital wallets, and high-ticket fraud detection mechanisms to secure premium transactions.
5. **Global Localization:** Introducing real-time currency conversion and multilingual support to cater to a broader, international demographic of watch enthusiasts.
6. **Enhanced Admin Analytics:** Upgrading the dashboard with predictive analytics to forecast sales trends, automated stock alerts, and bulk management tools for easier inventory handling.
7. **Customer Loyalty Programs:** Introducing a rewards system, referral bonuses, and a tier-based membership program for returning watch collectors to drive long-term retention.
8. **Automated Communication:** Implementing an automated sequence of email and SMS notifications for order tracking, wishlist price drops, and abandoned cart recovery.

Conclusion

The development of the **Watchera** platform successfully demonstrates a functional, full-stack e-commerce solution built using the MEAN stack (MongoDB, Express.js, Angular, and Node.js). The primary objective of creating a straightforward, responsive, and user-friendly online watch store was met effectively.

The system reliably handles core e-commerce workflows: users can seamlessly browse the catalog, manage their shopping carts, apply discount coupons, and place orders. Meanwhile, the administrative side provides essential controls for managing inventory, users, and order statuses. While designed as a minor project, Watchera establishes a solid, well-structured foundation. It successfully applies fundamental web development principles, database management, and API integration, resulting in a clean and efficient digital storefront that is fully capable of future scaling and feature additions.

References

<https://angular.io/docs>

<https://nodejs.org/en/docs/>

<https://expressjs.com/>

<https://www.mongodb.com/docs/manual/>

<https://tailwindcss.com/docs>

<https://developer.mozilla.org/>